

**AUSTRALIAN RESEARCH COUNCIL  
Discovery Early Career Researcher Award  
Proposal for Funding Commencing in 2021**

DE

**PROJECT ID: DE210100398**

**First Investigator: Dr Philipp Bayer**

**Admin Org: The University of Western Australia**

Total number of sheets contained in this Proposal: 45

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## Part A - Administrative Summary (DE210100398)

### A1. Application Title

(Provide a short title. (Up to 75 characters, approximately 10 words).)

The Life And Death Of Plant Genes

### A2. Person Participant Summary

(Add the DECRA candidate participating in this application.)

| Number | Name             | Participant Type                        | Current Organisation(s)             |
|--------|------------------|-----------------------------------------|-------------------------------------|
| 1      | Dr Philipp Bayer | Discovery Early Career Researcher Award | The University of Western Australia |

### A3. Organisation Participant Summary

(Please add all organisations participating in this application. Refer to the Instructions to Applicants for further information.)

| Number | Name                                | Participant Type           |
|--------|-------------------------------------|----------------------------|
| 1      | The University of Western Australia | Administering Organisation |

### A4. Application Summary

(Provide an Application Summary (which is used by the Minister to consider the application), focusing on the aims, significance, expected outcomes and benefits of this project. Write the Application Summary simply, clearly and in plain English. If the application is successful, the Application Summary will be used to give the general community an understanding of the research. Avoid the use of acronyms, quotation marks and upper case characters. Refer to the Instructions to Applicants for further information. (Up to 750 characters, approximately 100 words))

My recent work has demonstrated that in contrast to animal genes, many plant genes show presence/absence variation within a species, with associated trait variation. In this project, I will explore models of gene birth and death by comparing genomes of Brassicaceae, including the model Arabidopsis and Brassica crop species. By comparing many genomes I will learn how new genes were born. I will build models that predict the likelihood of gene loss based on a gene's physical environment, function, and expression. The project will build on our understanding of plant genetic diversity. Expected outcomes of this research include the identification of key genomic elements in gene birth and loss and support strategies to improve plant cultivars.

### A5. List the objectives of the proposed project

(List each objective separately by clicking 'add answer' to add the next objective. This information will be used for future reporting purposes if this application is funded. (Up to 500 characters, approximately 70 words per objective))

Objective

Identify genetic elements associated with 'young' genes and map the source of new genes

Objective

Map the genetic causes of gene loss in the Brassicaceae

Objective

Quantify how birth and death mechanisms differ between classes of genes

## A6. National Interest Test Statement

*(Outline the extent to which the research contributes to Australia's national interest through its potential to have economic, commercial, environmental, social or cultural benefits to the Australian community. Write the description of the national interest simply, clearly and in plain English between 750 characters and 1125 characters (between approximately 100 and 150 words).*

*Note: The National Interest Test Statement may also be used for public release by the ARC.*

)

Continued advances in DNA sequencing technology are revolutionising agriculture, however, our ability to interpret the vast quantities of data being generated has lagged behind our ability to generate the data. I have pioneered approaches to discover new insights into plant biology through the analysis of genome sequence data. Plant genomes are in constant flux with genes being added ('birth') and lost ('death'). While we know some of the mechanisms as to how genes are 'born' and 'die', we do not know how much each mechanism contributes to gene birth and death or how many additional factors are yet to be discovered. This project will address this limitation.

While this project focuses on fundamental science, the knowledge gained can be translated into practical applications in applied crop improvement. I have previously worked closely with breeding companies to ensure that discoveries can be used to help develop new crop varieties, supporting the Australian economy through food exports and rural communities who need improved varieties which can produce yield under an increasingly variable climate.

## Part B - Classifications and Other Statistical Information (DE210100398)

### B1. Does this application fall within one of the Science and Research Priorities?

|     |
|-----|
| Yes |
|-----|

| Science and Research Priority | Practical Research Challenge |
|-------------------------------|------------------------------|
| Food                          | Enhanced food production     |

### B2. Field of Research (FOR)

(Select up to three classification codes that relate to the DECRA candidate's application. Note that the percentages must total 100)

| Code                                                           | Percentage |
|----------------------------------------------------------------|------------|
| 060102 - Bioinformatics                                        | 50         |
| 060407 - Genome Structure and Regulation                       | 40         |
| 070305 - Crop and Pasture Improvement (Selection and Breeding) | 10         |

### B3. Socio-Economic Objective (SEO-08)

(Select up to three classification codes that relate to the DECRA candidate's application. Note that the percentages must total 100.)

| Code                                                                          | Percentage |
|-------------------------------------------------------------------------------|------------|
| 829999 - Plant Production and Plant Primary Products not elsewhere classified | 51         |
| 820502 - Canola                                                               | 49         |

### B4. Interdisciplinary Research

(This is a 'Yes' or 'No' question. If You select 'Yes' two additional questions will be enabled:

1. Specify the ways in which the research is interdisciplinary by selecting one or more of the options below.
2. Indicate the nature of the interdisciplinary research involved. (Up to 375 characters, approximately 50 words))

Does this application involve interdisciplinary research?

|     |
|-----|
| Yes |
|-----|

Specify the ways in which the research is interdisciplinary by selecting one or more of the options below.

|             |
|-------------|
| Methodology |
|-------------|

Indicate the nature of the interdisciplinary research involved. (Up to 375 characters, approximately 50 words)

|                                                                                                                                                                                                                    |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| The work combines genomics and bioinformatics, using national high performance computing infrastructure to understand genome diversity, with subsequent assessment and validation in the laboratory and the field. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### B5. Does the proposed research involve international collaboration?

(This is a 'Yes' or 'No' question. If You select 'Yes' two additional questions will be enabled:

1. Specify the nature of the proposed international collaboration by selecting one or more of the options below.
2. Specify the countries which are involved in the international collaboration.)

|     |
|-----|
| Yes |
|-----|

### B6. What is the nature of the proposed international collaboration activities?

(Select all options from the drop down list which apply to this application by clicking on the 'Add' button each time an option is selected.)

|                                                          |
|----------------------------------------------------------|
| Correspondence: eg email; telephone; or video-conference |
| Attendance at and/or hosting of workshop or conference   |

**B7. If the proposed research involves international collaboration, please specify the country/ies involved.**

(Commence typing in the search box and select from the drop-down list the name of the country/ies of collaborators who will be involved in the proposed project. Note that Australia is not to be listed and is not available to be selected from the drop-down list.)

|         |
|---------|
| Belgium |
| Germany |

**B8. How many PhD, Masters and Honours places will be filled as a result of this project?**

(For reporting purposes, the ARC is interested in reporting the number of Research Students that would be involved in this application if it is funded. Please enter the number of student places (full-time equivalent) that will be filled as a result of this project.)

Number of Research Student Places (FTE) - PhD

|   |
|---|
| 1 |
|---|

Number of Research Student Places (FTE) - Masters

|   |
|---|
| 4 |
|---|

Number of Research Student Places (FTE) - Honours

|   |
|---|
| 2 |
|---|

## Part C - Project Eligibility (DE210100398)

### C1. Medical Research

*(This is a 'Yes' or 'No' question. Does this project have content which requires a statement to demonstrate that it complies with the eligible research requirements set out in the ARC Medical Research Policy located on the ARC website?)*

No

### C2. Medical Research Statement

*(If applicable, justify why this project complies with the eligible research requirements set out in the ARC Medical Research Policy located on the ARC website. Eligibility will be based solely on the information contained in this application. This is the only chance to provide justification and the ARC will not seek further clarification. (Up to 750 characters, approximately 100 words))*

### C3. Current Funding

*(Does this application request funding for similar or linked research activities, infrastructure or a project previously funded, or currently being funded, with Australian Government funding (from ARC or elsewhere)? This is a 'Yes' or 'No' question. If 'Yes', provide the Project ID(s) and explain how funding this project would not duplicate Australian Government funding or overlap with existing projects.)*

No

If yes, provide the Funded Project ID

Detail how this project is different from the previously/currently funded project(s). (Up to 2000 characters, approximately 285 words)

### C4. Other Application(s) for funding

*(Are you applying for funding from the Australian Government (ARC or elsewhere) for similar or linked research? This is a 'Yes' or 'No' question. If 'Yes' provide the application ID(s) and briefly explain why more than one application for similar or linked research has been submitted.)*

No

If yes, provide the application ID

Briefly explain why more than one application for similar or linked research has been submitted (Up to 2000 Characters, approximately 285 words)

## **Part D - Project Description (DE210100398)**

### **D1. Project Description**

*(Upload a Project Description as detailed in the Instructions to Applicants and in the required format. Ensure that the Project Description responds to the Assessment Criteria listed in the grant guidelines. (Up to 10 A4 pages))*

Uploaded PDF file follows on next page.

## PROJECT TITLE

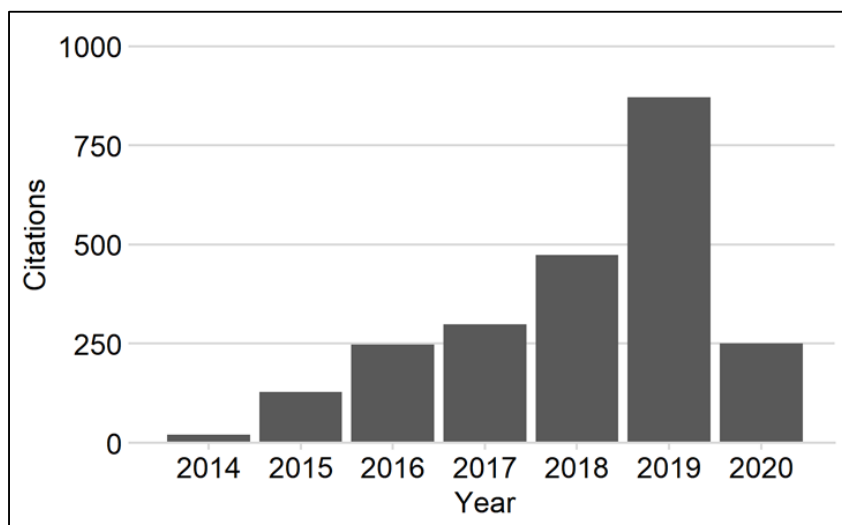
The life and death of plant genes - what drives new genes to arise and how are genes lost?

## INVESTIGATOR/CAPABILITY

During my undergraduate studies, I discovered a talent for computational analysis of biological data which led me to pursue an internship analysing evolutionary constraints in HIV genes, in 2008. It quickly became apparent that I wanted to pursue a career in applied bioinformatics. I therefore decided to undertake a Masters in IT to develop my computational skills and subsequently applied these to my PhD studies in Applied Bioinformatics, which I undertook in one of the few groups internationally which is 100% computational but focussed on practical outcomes for biology. During my PhD studies, I analysed genomic data from large populations of canola (*Brassica napus*), working with industry partners to assess genome assembly quality and performing genome-wide trait association studies. Through this, I discovered genomic variation which underlies the inheritance of important agronomic traits.

I extended the genome diversity analysis during my first postdoctoral position, where worked at the forefront of the emerging field of pan genomics, using big data computing skills to assemble pangenomes (a pangenome represents the entire gene content of the species, not just those in the reference genome) for several plant species including the very first plant pangenome *Brassica oleracea* (cabbage, kale, cauliflower), *Brassica napus* (canola), *Triticum aestivum* (wheat), *Amborella*, and *Cajanus cajan* (pigeon pea). Through this I characterised the abundant gene presence/absence variation within these species, training more than 10 PhD students in this and related areas. The large data analysis skills required in pangenome analysis are highly sought-after and these students move on to other leading international groups on submission of their thesis. One of the major findings of the pangenome work is that there is extensive gene presence/absence variation in plants, especially in disease resistance (*R*) genes. This presence/absence has a significant impact on the phenotypic diversity, and especially disease resistance diversity, commonly observed in plants. While my work answers questions in biology through understanding how the structure and diversity of plant genomes contribute to plant phenotypes, it also raises new questions, and in the case of this proposal, I am asking: what drives new genes to arise and how are genes lost?

This project involves comparative genomics on a very large scale. My background is in applied bioinformatics and comparative genomics and I have worked extensively on the comparative genomics of plant species and families, including Brassicaceae, diverse legumes, and cereals, work upon which this project will expand significantly. This project will provide insights into the mechanisms of gene birth and gene loss, mechanisms which are the basis of much of the plant diversity we see today. Understanding these mechanisms will enable plant breeders to generate novel genetic diversity while maintaining the conservation of agronomically important genes. The project will also support an extension of the established pangenomics and comparative genomics toolsets to apply machine learning and modelling methods. These methods will enable me to identify the complex mechanisms which lead to gene presence/absence variation in plants.



**Figure 1:** Citations for my publications per year as reported by Google Scholar ranging from 2014 to 2020. Data accessed on 3.3.2020.

My research performance is demonstrated by my publications. I have authored 43 papers in high impact and highly ranked journals including Science (2 publications, IF 41, ranked 2 out of 69 in JCR Category Multidisciplinary Sciences), Nature Genetics (1 publication, IF 25, 2/174 Genetics & Heredity), Nature Communications (1 publication, IF 11.8, 5/69 Multidisciplinary Sciences), Nature Plants (1 publication, IF 13.3, 3/228 Plant Sciences), New Phytologist (1 publication, IF: 7.3, 8/228 Plant Sciences), Plant Physiology (1 publication, IF: 6.3, 10/228 Plant Sciences), Plant Biotechnology Journal (6 publications, IF: 6.8, 9/228 Plant



Sciences), Trends in Genetics (1 publication, IF: 10.6, 7/174 Genetics & Heredity), Trends in Biotechnology (1 publication, IF: 13.7, 4/162 Biotechnology), Theoretical & Applied Genetics (3 publications, IF: 3.9, 6/89 Agronomy), and Frontiers in Plant Science (5 publications, IF: 4.1, 20/228 Plant Sciences). These publications have been cited more than 2,200 times since 2015, with rapidly increasing citation numbers each year (Figure 1) giving me a current H index of 19. Usually, H indexes for plant biologists grow by 1 every year. I am four years out of my PhD which means that my H index has been growing faster than expected. Much of my research has been on plant genome assembly, validation and analysis. As part of my PhD studies, I developed a novel way to genotype population-wide single nucleotide polymorphisms (SNPs). Using these SNPs, I identified precise recombinations within populations and applied this for high-resolution trait mapping with industry partners. The method also permitted the development of an approach to detect and correct errors in genome assemblies. I first applied this to assess the quality of the large, amphidiploid *Brassica napus* genome assembly, leading to a major improvement in accuracy and contiguity, and co-authorship on the resulting paper published in Science (IF: 41) [1]. I also applied this method to improve the quality of the highly repetitive pea genome, which was published in 2019 in Nature Genetics (IF: 25.5) [2]. Through my efforts to develop new approaches for genome assembly analysis and validation, I established a method which uses genomic reads sequence reads from single chromosomes, isolated using flow-cytometry, and applied this to improve the assembly of several plant species including wheat [3], chickpea [4], lentil (in preparation) and pea [2]. These genomes represent some of the largest, most complex and repeat-rich genomes published, and assembling them to such high quality was a major challenge, requiring innovative computational approaches. These genomes remain the standard reference for these species, even with recent advances in the quality of data produced by the latest genome sequencing systems.

As part of an international effort, I contributed to the assembly of the first wheat genome, a very large and repetitive genome (16 Gbp) [3]. This assembly used DNA cloned within bacterial artificial chromosomes (BACs) and short-read Illumina DNA sequencing. I developed an approach which used the BAC sequences to order the assembled contigs into a pseudomolecule. This required a significant amount of coding work on my part and enabled the scaffolding of the assembly pieces. This work was published in Science (IF: 41) [3] and has already been cited 382 times in less than two years since publication.

During my doctoral studies, I worked closely with the plant breeders at Bayer CropScience (Ghent, Belgium), now BASF. I also collaborate with KWS in Germany and Intergrain in Australia. Working with industry gives me access to large quantities of data, and I was one of the first researchers to have access to genome assemblies for multiple individuals of the same plant species. During comparative analysis, it quickly became apparent that there was far more gene presence/absence variation in plants than previously considered, and that this diversity was responsible for important agronomic traits. While I was unable to publish these initial discoveries which were made in commercial canola varieties in collaboration with an industry partner, the concepts led to the first plant pangenome study in *Brassica oleracea*, a genus that includes important vegetables such as cabbage, kale, and cauliflower [5]. In this project, I co-supervised a PhD student Agnieszka Golicz to design the assembly and comparison pipeline and co-authored the resulting paper in Nature Communications (IF: 11.8). Based on this experience I have since been invited to contribute to many other plant pangenome projects, including wheat [6], canola [7], pigeon pea [8], and a focussed analysis of 262 barley exome datasets for *restorer-of-fertility* genes [9]. I have searched the *B. oleracea* and the *B. napus* pangenome for diverse *R*-gene alleles which led to several newly discovered *R*-genes in both species [10, 11] which are currently being assessed for their resistance function. We are currently finalising manuscripts describing construction and analysis of a soybean pangenome which examines gene conservation and loss for more than 1000 individuals, representing the whole US soybean breeding program. We have completed the first banana pangenome identifying genetic differences both within and between *Musa* and *Ensete* banana, and expect to submit the manuscript in the coming weeks. I also study non-crop species of evolutionary interest and have co-authored a manuscript describing the pangenome of *Amborella*, the only remaining sister species to all other flowering plants. The manuscript is currently undergoing revision for submission to Nature Communications.

My recent work includes comparative genomics of disease resistance genes (*R*-genes) across 40 published Brassicaceae genomes. During this analysis, I discovered that *R*-gene numbers vary strongly between species and even between multiple assemblies of the same cultivar. This observation suggested an issue with assembly or annotation which I pinpointed to errors in a standard repeat masking database used across plant species and which led to the erroneous masking of real genes. This discovery was published in Nature Plants where I am first author [12]. Current projects include developing a map of novel *R*-genes which are

absent from commercially grown Brassicaceae and may be of interest to breeders. In all of these projects, I play a lead role, supervising students, developing and applying the assembly pipelines, comparing and validating the assemblies, and streamlining each of the computational steps to ensure efficient use of high performance compute time. Another current project examines gene conservation and loss at the species level following polyploidy by comparing the pangenomes of *B. napus* and its diploid relatives, *B. rapa* and *B. oleracea*. I have assembled three *Brassica* pangenomes using data from more than 300 *B. rapa*, *B. oleracea*, and *B. napus* individuals, and used these to map differences in gene conservation and loss across the three species and to identify novel disease resistance genes. I am the lead researcher in this project and have performed the majority of the analysis. The main outcomes of this project are a map of gene loss following polyploidy in *B. napus*, and the proposed DECRA project builds on these results.

## PROJECT QUALITY AND INNOVATION

**Significance:** There is a major investment in genome sequencing projects globally, yet capitalisation on this investment is limited by the lack of experienced genome informatics researchers who combine a solid foundation in biology with the ability to manage and analyse the quantities of data being generated. This project will develop capability in genome analysis and applied bioinformatics to understand the foundations of genome evolution. Given the growth of the field of genomics, training will be an important aspect of this project, both extending my skills and experience. In this project, I propose to train one PhD, 4 Masters and 2 Honours students in applied bioinformatics.

Bioinformatics and genomics skills are in extremely high demand, yet at the same time, there is very little training occurring for bioinformatics in Western Australian universities. Training PhD, Masters and Honours students via this proposal closes a significant training gap in Western Australia. The PhD-student will work across all three aims outlined here from genome assemblies, to comparative genomics, to the critical evaluation of machine learning models. The Masters and Honours students will work on smaller, tightly defined projects based on results the PhD student and I have discovered.

Climate change is an ongoing threat to Australian agriculture, with predictions of reduced rainfall [13] and rising temperatures. Combined with a growing global population and the associated demand for Australian crop exports, there is an urgent need to breed plant cultivars with greater resilience to biotic and abiotic stress [14]. Canola is the third most commonly grown crop in Australia worth 2 billion dollars each year. Canola is part of the diverse and economically important Brassicaceae family, which also contains *B. oleracea* (broccoli, cabbage, kale), *B. juncea* (vegetable mustard), *Raphanus sativus* (radish) and the model plant *Arabidopsis thaliana* (thale cress). This project proposes to understand how novel genes arise and how genes are lost across the Brassicaceae. Understanding how genes are formed and lost will allow us to identify and introduce increased diversity into crops. A greater understanding of gene birth and death will allow us to understand how evolutionary processes have led to the crop genomes we have today and support the breeding of improved varieties. The project builds on my previous collaborations with industry to apply diversity to accelerate the production of new crop varieties, and will subsequently benefit Australian agricultural and rural communities through more resilient crops.

### Aim 1: Identify genetic elements associated with 'young' genes and map the source of new genes

I propose to use comparative genomics on a large scale, by sampling and comparing all published Brassicaceae genomes (currently >50). This will expand on my comparison of disease resistance genes by comparing all classes of genes. Whilst similar work has previously been carried out on a smaller scale [15] where only *Arabidopsis thaliana* and *A. lyrata* were compared, no origin could be assigned for roughly 35% of lineage-specific genes. Here I will expand on this approach. First, by comparing all Brassicaceae sets of genes with each other I will identify sets of lineage-specific genes for all lineages within the Brassicaceae. Sequencing technology has an impact on the number of genes observed in a genome assembly, as duplicated regions often collapse into single copies in short-read genome assemblies, and these differences may skew the outcomes of the comparative genomics pipeline. Therefore, I will re-assemble older Brassicaceae short-read assemblies using long-read sequencing technologies, as produced by the Oxford Nanopore PromethION. I will also assemble yet unassembled genomes of wild *Brassica* relatives *B. villosa*, *B. rupestris*, *B. insularis* and *B. montana*. These are *Brassica* relatives which have been sequenced by international collaborators before for their interesting disease resistance properties, but the coverage was too low to assemble these genomes well. Table 1 presents a list of the wild *Brassica* species to be assembled as well as the Brassicaceae species to be reassembled along with the reference of their original publication.

These species represent species from both Brassicaceae lineages which have been assembled only using Illumina data.

To resequence these genomes, I will use the assembly and polishing software pipeline Flye [16] + 4 rounds of RACON [17] polishing with available Illumina data, followed by Medaka polishing with Oxford Nanopore PromethION data to generate high-quality contiguous assemblies. Genome annotation will be performed using available RNAseq. This is an ideal chance for the PhD student to learn how genome assembly and gene prediction algorithms work and how to critically assess their results in the presence of false positive and false negative gene predictions. It will also be an opportunity for the student to learn how to use high-performance computing, especially high memory nodes, as high memory is required by genome assembly algorithms. Expertise in the usage of high performance computing is a valuable skill

As part of aim 1, I will:

1. Compare the position of protein-coding genes with predicted transposable elements (TEs). TE exaptation is one of the common sources of novel genes [18]. There is extensive evidence for plants 'domesticating' TEs as exapted TEs can confer phenotypic benefits. For example, the *FHY3* family of plant transcription factors is related to DNA *MuDR* family transposons [19]. These transcription factors confer phenotypic advantage as they are linked with increased responsiveness to far-red light. I expect to find many more similar TE exaptations in the Brassicaceae. TEs will be predicted using the *de novo* TE prediction pipeline EDTA [20]. I will then use sequence comparison software like BLAST+ [21] and protein domain comparison software like Interproscan [22] to compare the sequences of *de novo* predicted TEs with predicted genes to find genes similar to TEs, thereby identifying genes that are candidates for full (entire sequence) or partial TE exaptation. I will perform this analysis on all available Brassicaceae genome assemblies to comprehensively identify genes that are partially or completely based on TEs.
2. Discover long non-coding RNAs (lncRNAs) in all Brassicaceae genomes as these and similar genomic elements have been associated with novel genes [23]. Some lncRNAs have a sequence very similar to transposable elements, indicating a role of TEs in the creation of lncRNAs [24]. lncRNAs, in turn, play a role in the birth of some genes as they show similar coding potential and sequence constraints as young proteins, indicating that there is a hitherto unknown mechanism transforming lncRNAs into 'real' genes. [23]. The vast majority of lncRNAs are lineage-specific which makes them an ideal target of large-scale comparative genomics. As with TEs in step 1, I will use sequence comparison to compare the predicted lncRNAs with all predicted genes to search for lncRNA to gene overlap, thereby identifying genes which are likely to originate in lncRNAs. I will search all Brassicaceae annotations using established pipelines comparing predicted genes with miRNA and lncRNA databases [25].
3. Identify non-syntenic genomic regions by performing all-vs-all genomic comparisons using MCScanX software [26] between closely related Brassicaceae species. Non-syntenic regions between different *A. thaliana* cultivars have different rates of evolution and are enriched for disease resistance genes [27]. These regions likely arose due to tandem duplication with the potential for subsequent neo- and sub-functionalisation. Genes within these regions are therefore likely to have arisen due to local tandem duplications. I will use local realignments using BLAST+ [21] to search the non-syntenic region in surrounding areas of the same genomes to confirm that this region is indeed due to a tandem duplication and not due to an error in the genome assembly.

**Table 1:** Illumina-only Brassicaceae genome assemblies that will be reassembled or assembled for the first time

| Genome to be assembled       | Reassembled or novel assembly |
|------------------------------|-------------------------------|
| <i>Arabis alpina</i>         | Reassembly                    |
| <i>Aethionema arabicum</i>   | Reassembly                    |
| <i>Brassica insularis</i>    | Novel                         |
| <i>Brassica montana</i>      | Novel                         |
| <i>Brassica rupestris</i>    | Novel                         |
| <i>Brassica villosa</i>      | Novel                         |
| <i>Cardamine hirsuta</i>     | Reassembly                    |
| <i>Lepidium meyenii</i>      | Reassembly                    |
| <i>Raphanus raphanistrum</i> | Reassembly                    |
| <i>Raphanus sativus</i>      | Reassembly                    |
| <i>Tarenaya hassleriana</i>  | Reassembly                    |
| <i>Thlaspi arvense</i>       | Reassembly                    |

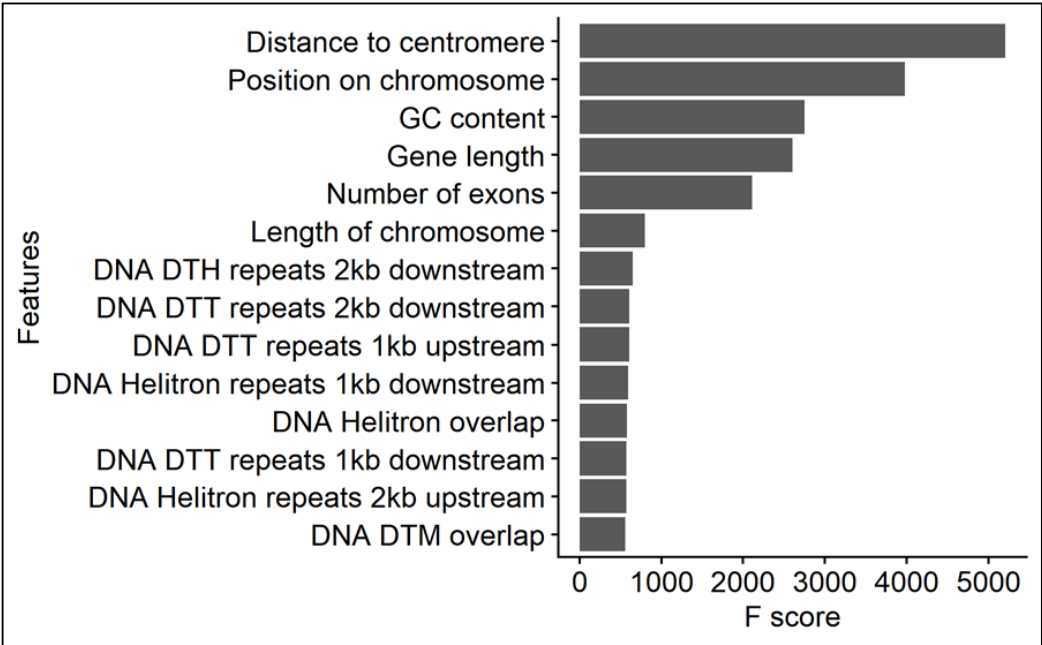
4. Perform whole-genome alignments for all Brassicaceae genomes within each lineage. This will allow me to map, for each pair of genomes, whether novel genes have arisen through frameshift mutations, which genes are directly overlapping gene models, and which genes have only intergenic alignments in the other genome, indicating a novel gene. Standard alignment approaches can lead to false-negative alignments and therefore supposed lineage-specific genes that in reality are shared between lineages, which can be fixed by using software such as abSENSE [28], which I will employ in this step to reduce false lineage-specific genes.

The combination of this analysis will allow me to assign sources for most, if not all, lineage-specific genes within the Brassicaceae. This will lead to the first Brassicaceae-wide study detailing exactly how each of the roughly 2 million Brassicaceae-genes arose in the past, and which gene is specific to which lineage. The results of this study will allow me to assign ages to lineage-specific genes, as genes occurring only within a specific lineage may only be as old as that lineage. The ages of all Brassicaceae lineages have previously been estimated [29]. I will build models linking ages as well as known genomic events such as transposon amplification, gene and whole-genome duplication to novel genes, on a genome-wide and gene-by-gene basis.

### Aim 2: Map the genetic causes of gene loss in the Brassicaceae

I have recently finished assembling *Brassica* pangenomes representing over 300 lines from *B. napus*, *B. rapa*, and *B. oleracea*. Within these lines, I have identified variable genes and assessed their position in the genome compared to different classes of transposable elements, the ends of the chromosomes and their distance from centromeres. I then used advanced machine learning techniques, such as XGBoost [30], in combination with Shapley values [31] to understand which mechanisms are associated with gene loss. These techniques work by changing each class of input data to the model to see how the prediction of the model changes. If the prediction changes strongly, the class of the input data is likely to have a strong impact in reality.

My preliminary results suggest that in the polyploid *B. napus*, the position of the gene on the chromosome has a much stronger impact on gene variability than association with any class of transposable elements (TEs) (Figure 2), though this is not true in the diploid progenitors *B. rapa* and *B. oleracea* which demonstrated a greater association with specific classes of TEs (unpublished). This suggests that different mechanisms dominate for gene birth/death between the diploid and polyploid crop species. In this aim, I will expand on these results by building more complex models for a wider range of Brassicaceae species.

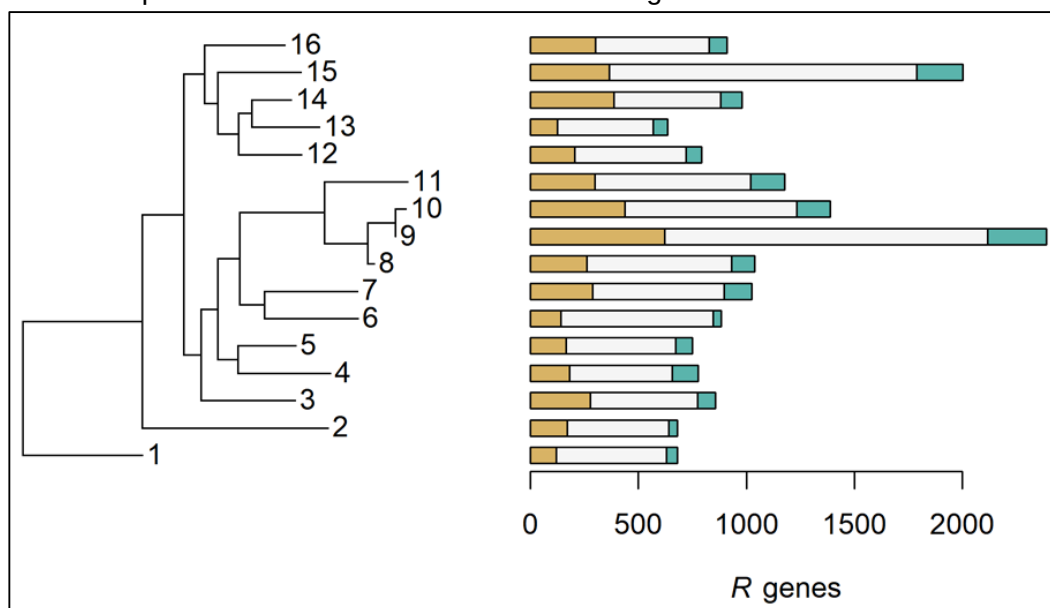


**Figure 2:** Association of genetic elements in the vicinity of variable genes in *B. rapa*. Features are sorted by importance with most important features on top. The higher the F score (x-axis), the stronger the impact on prediction of gene variability. In this case, ‘Distance to centromere’ is the feature with the strongest impact on gene variability prediction. Only a portion of the results are presented due to space constraints. Features with weaker impacts on gene variability predictions (F score below 900) are all linked with different superfamilies of DNA (class 2) transposons: DNA DTH (*PIF/Harbinger/Tourist*-like TEs), DNA DTT (*Tc1/Mariner/Stowaway*-like TEs), DNA DTM (*Mutator/Mule*-like TEs) and DNA Helitron superfamilies.

So far I have only used the state-of-the-art XGBoost library to use boosted trees for prediction of variable genes. There are many more machine learning techniques I have not been able to use, due to time constraints. As part of this project, the PhD student will perform a comprehensive evaluation of machine learning methods to see which model performs best. This will be extended, by the PhD student, using the technique of ensembling, where several different predicting machine learning methods are 'ensembled' into a single predictor. The student will also learn how to evaluate and implement techniques for the interpretation of machine learning models. So far I have only employed Shapley values to learn what the model has learned, but there are many more methods such as Global and Local Surrogate models that I have not yet employed. The student will work on evaluating these different methodologies. This will teach the student how to critically evaluate machine learning methods, the training process and the models' predictions.

To build these models I will need gene presence/absence tables which are typically built as part of standard pangenome assembly pipelines. I will assemble additional pangenomes based on available data to assess whether the gene variability patterns described above are similar in other Brassicaceae. As part of the 1,001 *Arabidopsis* genome initiative [32] there is currently data available for 435 lines which I will use to build an *Arabidopsis* pangenome.

There are currently no pangenomes for the BB and AABB genomes of *Brassica* (*B. nigra* and *B. juncea*), however, there is abundant public data for both species available within the Sequence Read Archive (SRA), therefore I will also build pangenomes for *B. juncea* and *B. nigra*. At the time of writing (March 2020) there are 294 libraries available for *B. juncea* (AABB) and 47 libraries available for *B. nigra* (BB). The Batley lab has additionally sequenced 273 *B. juncea* individuals as part of other, ongoing projects which will be used for this *B. juncea* pangenome. I will then compare these pangenomes with my pangenomes for *B. napus* (AACC), *B. oleracea* (CC), and *B. rapa* (AA). I will then extend these *Brassica* pangenomes by including data for available wild *Brassica* species. There is publicly available SRA data for multiple wild individuals of *B. villosa*, *B. rupestris*, *B. carinata*, *B. insularis*, *B. montana*, and *B. cretica*. As part of Aim 1, I will also generate sequencing data and assemblies for *B. villosa*, *B. rupestris*, *B. insularis* and *B. montana*. By including these wild species in the *Brassica* pangenome I will be able to identify and characterise genetic diversity that has not been introduced into domesticated *Brassica* species. This is especially important as these wild plants often contain disease resistance genes and alleles not found in domesticated plants.



**Figure 3:** Predicted *R*-genes for 16 Brassicaceae species compared with Brassicaceae-wide phylogeny as reported by Orthofinder. Brown: NLR genes (NBS-LRR domain present), grey: RLK genes (receptor-like kinase domain present), green: RLP genes (receptor-like protein domains). Only a subset of available Brassicaceae species shown and species names were replaced by numbers to save space. 1: *Tarenaya hassleriana*, 2: *Aethionema arabicum*, 3: *Arabis alpina*, 4: *Thlaspi arvense*, 5: *Eutrema salsugineum*, 6: *Sisymbrium irio*, 7: *Schrenkiella parvula*, 8: *Brassica rapa*, 9: *Brassica napus*, 10: *Brassica oleracea*, 11: *Raphanus sativus*, 12: *Arabidopsis thaliana*, 13: *Capsella grandiflora*, 14: *Boechera stricta*, 15: *Lepidium meyenii*, 16: *Cardamine hirsuta*.

### Aim 3: Quantify how birth and death mechanisms differ between classes of genes

In our previous studies, constructing and characterising the pangenomes of wheat [6], *B. oleracea* [5, 10], *B. napus* [7], sesame [33] pigeon pea [8], soybean, chickpea and banana, we observed that plant disease resistance genes (*R*-genes) are consistently among the most variable gene classes. *R*-genes can be divided into three classes – NLR (containing NBS-LRR domains), RLKs (containing receptor-like kinase domains), and RLPs (containing

receptor-like protein domains). Conversely, housekeeping genes involved in fundamental cellular processes are among the most invariable gene classes. *R*-genes are under selection to increase variability through the generation of new genes and genetic changes in existing genes, as they are in an evolutionary arms race with pathogens. Conversely, housekeeping genes are under negative selection, as any changes within these genes may lead to loss of fitness. This aim will apply the knowledge obtained in Aims 1 and 2 to learn how gene classes with different predicted functions, such as *R*-genes and housekeeping genes, generate diversity and maintain it on a genomic level.

I have preliminary results for *R*-genes from all available Brassicaceae (Figure 3) showing the diversity in the total number of *R*-genes across the Brassicaceae. Within the different Brassicaceae genome assemblies the total number of *R*-genes ranges from 635 to 2,391, mostly due to different ploidy levels and genome sizes, but also due to tandem duplications and localised gene variability. I will extend on these preliminary results by assessing *R*-gene annotations for newly released assemblies, as well as my additional Brassicaceae assemblies. This is an opportunity for the PhD student to

Once I have built and compared the preliminary models described above, I will extend them using more complex data. Currently, my models account only for physical position and co-location with genetic features but the model does not account for the relative importance or function of the gene within the organism. The model does not know whether the gene is essential for the survival of the plant, the plant's level of expressed genes across diverse tissues or its predicted function. For many of the published Brassicaceae genomes, such as *Arabidopsis* and *B. napus*, extensive gene expression data is available. I will use these data to classify genes based on their expression level and expression diversity across treatments and tissues. Genes that are lowly or rarely expressed may be at greater risk of being lost, as the organism may depend less on these genes. However, some rarely expressed genes may be important in specific environments such as abiotic or abiotic stress and may, therefore, be highly conserved. This improved model will help identify the relative importance of genes which may be associated with gene variability.

I will extend my current models by comparison of all genes with the well-annotated model plant *A. thaliana* to assign genes into functional classes. Furthermore, I will assess the interactions between genetic elements in the model. For example, some classes of transposons capture and copy genes to a new location, leading to duplicate copies which may undergo neo- or sub-functionalisation, with the potential for the production of hybrid genes and novel gene promoters. Currently, my models only check for the presence of single transposons of any class in the vicinity. Here I will extend the models to identify interactions between transposons of different classes as well as other genetic elements. I will then build different models for different classes of genes. I will start by comparing *R*-genes and housekeeping genes but will expand to other classes such as the most common class of genes encoding pentatricopeptide repeats (PPRs), or genes of agronomic interest such as stress or flowering time-related genes. These models will tell me how patterns of gene loss differ between these classes of genes.

## **BENEFIT**

The 'big data revolution' has arrived in biology with modern sequencing machines being able to generate several gigabytes of data per day. New genome assemblies, even for complex plant genomes, are being released weekly, and soon daily. This has led to a flood of data that needs to be analysed, but the skills and methods for efficient analysis of these very large sets of genomes are still being developed. The proposed DECRA project will result in newly acquired skills and novel methods which I will apply and teach the PhD student, honours and Masters students as part of this project.

This project focuses on gene birth and death in the Brassicaceae, which includes canola (*Brassica napus*), an important Australian crop. Canola is the third most important crop in Australia, with more than 60% of all canola grown in Western Australia [34]. Averaging the five years up to 2018-19, Australia produced a yearly average of 3.6 million tonnes of canola with up to 75% being exported. This makes Australia one of the leading exporters of canola in the world. At the same time, canola crop yields are already being impacted by climate change. This means that Australian growers will need novel cultivars adapted to these extreme conditions faster.

This project will result in several new high-quality polished long-read genome assemblies for the Brassicaceae, including several high-quality genomes for wild-type *Brassica* species which have never been assembled before. The wild-type genomes will form a valuable resource for other plant researchers as they



contain gene content and genetic variants that are not contained in other genomes. The high-quality reassemblies of already published genomes will serve as improved references for plant genomics researcher. The higher quality of these assemblies compared to the originally published assemblies means that these assemblies contain better representations of the true genetic and transposable element content of these species' genomes. This project will result in new *Brassica*-wide pangenomes including the A, B, and C genomes along with wild relatives. These pangenomes contain novel disease resistance genes which are of special interest to plant breeders as disease resistance is one of the main concerns of plant breeders. Which plants contain exactly which *R*-genes will be valuable to breeders as this will inform breeding programs as to which cultivars and wild relatives should be used as a source of novel *R*-genes into elite cultivars.

This project will result in new knowledge across the life-cycle of genes. There has been a flood of genomic data in recent years and I propose a project which will tie together disparate datasets and different genome assemblies to learn more about the fundamental mechanics of gene birth and death in plants. By learning how genes arise in Brassicaceae we will be able to explain where certain classes of genes come from, which will give us evolutionary insights into which genetic elements are associated with new genes. This project will result in knowledge of the mechanics of gene birth and death. In the long run, the outcomes of this project will allow plant breeders to manipulate the *de novo* gene birth process to generate new genes or alleles of interest.

## FEASIBILITY

I have significant experience in comparative genomics, genome assembly, and pangenomics as evidenced by my work on the canola genome [1], genotyping-by-sequencing in canola and chickpea [4], the first plant pangenome in *Brassica oleracea* [5], the wheat pangenome [6], the *Brassica napus* pangenome [7], the first complete wheat genome assembly [3], investigating biases in disease resistance gene prediction in *Brassica napus* [12], comparative discovery of *restorer-of-fertility* genes in barley [9], two studies on discovering novel disease resistance genes in the *Brassica oleracea* [10] and *Brassica napus* [11] pangenomes, the pea reference genome [2], and the pigeon pea pangenome [8]. Based on these publications I propose this project, which will result in additional Brassicaceae genome assemblies and pangenomes, but then extends on these datasets by building a comprehensive model of gene birth and death across the Brassicaceae. This involves discovering genetic elements such as transposable elements, long non-coding RNAs, non-syntenic regions along with whole-genome alignments across all Brassicaceae.

This project is highly feasible as preliminary data has already been produced for aims 1 and 2 (Figure 2, Figure 3). Due to these preparations and available public data, I expect to achieve rapid results including finished genome assemblies, subsets of assembled pangenomes, and sets of predicted transposable elements (Table 2). As evidenced by my many publications I have extensive experience in comparative genomics, pangenomics, the discovery of genetic elements, critical interpretation of genome assembly and gene predictions, and gene presence/absence calling. I have extensive experience in using large supercomputing clusters due to my many years of using the Pawsey Supercomputing Centre which enable to efficiently carry out computational tasks with large amounts of data in the petabyte range. All these skills are required in the proposed project making me the ideal candidate for it.

**Table 2:** Timeline for genome assembly production, genetic element discovery, and model building over three years

| Aim | Activity                                                            | 2021 | 2022 | 2023 |
|-----|---------------------------------------------------------------------|------|------|------|
| 1   | Collection of Brassicaceae genome assemblies                        |      |      |      |
| 1   | Perform all-vs-all Brassicaceae genome assembly comparisons         |      |      |      |
| 1   | Assemble additional Brassicaceae genomes                            |      |      |      |
| 1   | Estimate gene age based on which lineages contain which genes       |      |      |      |
| 1   | Discover genetic elements in all collected genomes                  |      |      |      |
| 1   | Map most likely source for each Brassicaceae gene                   |      |      |      |
| 2   | Build additional <i>B. juncea</i> and <i>A. thaliana</i> pangenomes |      |      |      |
| 2   | Build models predicting and explaining gene variability             |      |      |      |
| 3   | Classify all genes into <i>R</i> -genes and house-keeping genes     |      |      |      |
| 3   | Quantify how gene birth/death mechanisms differ in subclasses       |      |      |      |
| 3   | Build different models for different subclasses of genes            |      |      |      |

Aim 1 is highly feasible as similar work has been carried out on a much smaller scale in the Brassicaceae using two genome assemblies [15]. While expanding this work using all available Brassicaceae genomes will certainly take more time and computational resources, the approach itself will not have to be substantially modified. However, due to different genome assemblies and annotations, this project will require a major effort in cleaning and normalising genome assemblies and annotations to have comparable sets of genes. I have extensive experience assembling, annotating and validating plant genome assemblies. I also have extensive experience in carrying out large-scale comparative genomics with multiple pangenomes, all demonstrated by multiple high impact publications.

Aim 2 is highly feasible as I have already gathered preliminary data and results (Figure 2). I am also in constant contact and exchange with other plant comparative genomics researchers whose expertise will contribute to this project. This aim requires machine learning knowledge, an area of expertise I have gained increasing experience of over the last four years. I am a member of the Perth Machine Learning Group, an industry-heavy weekly meet-up of machine learning practitioners. I also co-supervise two PhD students jointly with Prof. Mohammed Bennamoun, from the computer science department at the University of Western Australia, in projects applying machine learning to crop genomic and image data. Prof. Bennamoun is a world-leading authority on machine learning applications.

The success of aim 3 depends on the success of aim 1 and aim 2. As I have shown above, these two aims are highly feasible. I have preliminary results and established pipelines to discover gene classes of interest in all Brassicaceae genomes. I have already finished a preliminary analysis of disease resistance gene discovery for a subset of all Brassicaceae genomes (Figure 3). I have wide experience performing analyses with dozens or even hundreds of genomes. Aim 3 will require to search all Brassicaceae genomes for other classes of genes such as PPR genes or housekeeping genes. For this, I will need to run different pipelines, but on an identical computational scale to the *R*-gene discovery pipelines, I have used before. I am therefore confident that aim 3 is feasible.

This project is computationally intensive. I am in regular contact with the Pawsey Supercomputing Centre, as our group is one of the largest, if not the largest user of national high-performance computing infrastructure for computational genomics in Australia. I, therefore, have extensive experience in efficiently using computing on a very large scale to process the large amounts of data described here, and I have carried out similar workflows with related datasets. CPU-intensive jobs such as whole-genome alignments will be carried out on the Magnus cluster. I have started to move to GPU-based systems to replace many of the CPU-heavy steps of my pangenome workflows. For example, the GPU-based *nvBowtie*, a replacement of *bowtie2*, is up to ten times faster. During this project, I will continue to transfer my CPU-based workflows onto the new Pawsey GPU-cluster *Topaz*, which will allow me to run enormous job tasks in just a few days as GPU-based approaches are generally 5 to 10 times faster than standard CPU approaches. *Topaz* is a new GPU cluster with twenty GPU computing nodes that will receive additional nodes over the next few years. *Magnus* and *Topaz* represent some of the most powerful computing systems in the world which will ensure that the computationally intensive parts of this project will finish on time.

This project requires the storage of large amounts of data on the scale of hundreds of terabytes. I will use the Pawsey Supercomputing Centre's data storage facilities for this data. The */scratch* file system, which is intended for use during analysis, has 3.1 petabytes of space, which is abundant for the scale of this project. For long-term storage, *data@pawsey* is a tape-drive facility with essentially unlimited space, as additional tapes are continuously bought and connected. For secondary backups I will use the data storage facilities of the UWA library.

## COMMUNICATION OF RESULTS

I have worked for several years in the international plant genomics community and maintain excellent links with other researchers in this area across the world. These include many formal and informal points of contacts ranging from conference meetings and workshops to regular social media interactions. I will present my research at national and international conferences. Since the start of my scientific career, I have attended and presented almost every year at the Plant and Animal Genome (PAG) conference in San Diego, USA, the largest conference in genomics of plant and animal genomes, including pangenomics and comparative genomics. This conference attracts almost 4,000 visitors every year including most if not all of the leaders



of all non-human genomics fields. I will present results stemming from this DECRA project at PAG together with the requested PhD student.

During my career, I have established a strong social media presence. I will share progress on this project via my social media channels on Twitter (3,346 followers on @PhilippBayer, 430 on @BiolSci\_UWA) and Facebook (UWA School of Biological Sciences, 845 likes) where I receive regular positive feedback from international experts in the field. I will also publish data and code on Github and other representative repositories.

Due to my capability and expertise in this area, it is expected that outputs will be rapid and the results published in high impact peer-reviewed international journals. I have published in reputable plant genomics journals such as Nature Communications, New Phytologist or Plant Biotechnology Journal. I will publish in Open Access journals as required by the ARC. If the chosen journal does not support an Open Access option I will make an author-archived version available within UWA's Research Repository. Just as with my previous research, open data and code associated with these publications will enable the scientific community to reproduce my research within my own and the Applied Bioinformatics group's Github accounts. Along with the research and code I will also publish reproducible pipelines implemented in Snakemake.

Since the research presented here is of great interest to the Australian agricultural food sector we will expand and start new collaborations with industry. This will be carried out using established connections that our group already has in place with plant breeding companies such as BASF (Belgium), KWS (Germany), and Intergrain (Australia). I will continuously communicate our results to our plant breeding industry collaborators so that my insights can be directly transferred into an improved breeding process.

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## **D2. Statement by the Administering Organisation**

*(Provide a Statement that addresses the relevant criteria as set out in the grant guidelines. The Statement must be signed by the Deputy Vice-Chancellor (Research) or equivalent. (Upload a PDF of up to three A4 pages))*

Uploaded PDF file follows on next page.

## **D2. Strategic statement in respect of the project research environment**

The University of Western Australia (UWA) strongly supports Dr Philipp Bayer's application for a DECRA fellowship. Dr Bayer is a high-calibre early career researcher in the area of plant bioinformatics and we strongly endorse the proposed DECRA entitled "The Life And Death Of Plant Genes". Dr Bayer will be well-positioned at UWA within a network of plant bioinformatics experts who will mentor him and help propel him towards his already highly successful and independent career. In particular, Dr Bayer will be mentored by Professor David Edwards, who is an expert in plant bioinformatics, and Professor Jacqueline Batley, an expert in plant genetics and genomics.

### **Alignment of the DECRA project with core and emerging research strengths at The University of Western Australia**

UWA is one of Australia's leading research-intensive universities and ranked 99 in the Academic Ranking of World Universities (ARWU, 2019). Of particular relevance to this DECRA application, UWA is ranked 24 in Ecology (ARWU, 2019), 35 in Biological Sciences (ARWU, 2019) and 51-100 in Biological Sciences (QS World University Rankings, 2019). In the Excellence in Research for Australia 2018 assessment, UWA was rated well above world standard in Biological Sciences and well above world standard for Plant Biology. UWA was also highly rated for impact within the discipline of Biological Sciences in the Engagement and Impact Assessment (EIA 2018-19) for its translation of research beyond academia into the beneficial impact upon society, the economy and/or the environment. Data science and the analysis of "big data" have also increased in prominence and capacity at UWA over the last 5 years, with ready access to supercomputing capabilities through the Pawsey Centre.

Dr Bayer's DECRA is well aligned with the research priorities and strengths in agricultural science and plant biology at UWA. This project will contribute to help plant breeders introduce genes into new plant cultivars that remain stable over many generations. The project will also teach us how novel genes arise with the eventual aim to manipulate the gene birth pathway in order to create new genes without the need for exchange of genetic material. Consequently, this DECRA will contribute to a core focus within the Faculty of Science on crop genetics and breeding. The DECRA candidate's proposed research will lead to higher stability of economically important traits in crop plants, which broadly will provide benefits and security in food production, not just within Australia but also globally.

The DECRA candidate will be embedded in the Applied Bioinformatics group in the School of Biological Sciences at UWA, under the mentorship of Professor Dave Edwards and Professor Jacqueline Batley. Professor Edwards is a world-renowned bioinformatics researcher in the plant community and has a wealth of experience and knowledge to contribute to this project. Professor Edwards is a member of several international and national consortia: the Wheat Information System (WheatIS) and the Multinational Brassica Genome Project Steering Committee (MGBP). Professor Edwards is also a 2019 highly-cited researcher, with many publications in high-impact journals such as seven publications in Science, two in Nature Genetics, two in Nature Communications, and two in

Nature Plants. Professor Jacqueline Batley has made significant breakthroughs in disease resistance in the oilseed crop canola. She researches crop genetics to enable breeders to produce better crops with resistance to disease and climatic conditions. Professor Batley was awarded the prestigious 2019 Nancy Millis Medal for Women in Science, which recognises mid-career female scientists who have demonstrated exceptional leadership and established an independent research program in the natural sciences.

### **Support for the DECRA candidate**

The University strongly supports Dr Bayer. UWA is providing a total cash contribution of \$418,772, which represents a substantial contribution and demonstrates significant institutional commitment to support Dr Bayer in conducting the proposed research.

UWA fosters the development of its early career researchers through training programs, awards and funding opportunities. The University offers funding via a number of internal grant schemes, including Research Impact Grants (\$20,000) to fund activities to assist with promoting academic and non-academic impact of the candidate's research and Research Collaboration Awards (\$30,000) which have a dual purpose of promoting either international research collaboration or industry research partnerships. UWA will also provide cash contributions towards Dr Bayer's DECRA, including a salary supplement totalling \$143,522.

Other cash contributions towards the project include:

- UWA funded PhD stipend, \$105,000
- Graduate Research School travel funds, \$1,850
- PhD student workstations, \$1,500
- UWA PhD student research funds, \$4,500
- Scholarship for International Research Fees (SIRF), \$162,400

UWA will also provide considerable in-kind contribution in the form of access to the Pawsey Supercomputing Centre and whatever access is required to high quality molecular laboratories.

### **Pathway to a successful academic career**

The award of a DECRA based in the Biological Sciences at UWA will see Dr Bayer welcomed into a strong collaborative, intellectually enriching and academically outstanding environment. Dr Bayer will be offered opportunities to be actively involved in the School's teaching program, not exceeding 20% of his FTE. Dr Bayer has supervised 10 HDR students and provides extensive advice and supervision through his own mentoring initiatives to other PhD, Masters and undergraduate students. He will be given opportunities to further develop in this capacity throughout his DECRA, including formal training in supervision. The University recognises that the skills and qualities gained from opportunities to teach, supervise and mentor are essential for academic development and career progression.

The University actively pursues on-going appointments for successful DECRA recipients based on performance during the fellowship, alignment with the strategic needs of the University and Head of School approval. To foster this, Dr Bayer will be given every opportunity to advance his career and establish himself as an independent researcher at UWA. Dr Bayer will have access to career progression opportunities through UWA's Organisational Development division and the Centre for Education Futures, which provide resources to enhance teaching and learning as well as professional and leadership training programs. Moreover, the proposed research project is well-designed and will likely result in vital findings to advance our knowledge of plant bioinformatics and provide important benefits when applied in the future. These outcomes will advance the development of Dr Bayer and capability in plant genomics and crop breeding in Australia. With the outstanding research environment and mentoring opportunities available at UWA, and within the School of Biological Sciences in particular, this DECRA fellowship will further solidify the academic career of Dr Bayer and grow his reputation as an expert in plant bioinformatics.

It is a pleasure to recommend Dr Bayer for this DECRA fellowship.



**Professor Tim Colmer**  
**Deputy Vice-Chancellor (Research)**

## Part E - Project Cost (DE210100398)

### E1. What is the proposed budget for the project?

(There are rules around what funds can be requested from the ARC. You must adhere to the scheme specific requirements listed in the grant guidelines. Refer to the Instructions to Applicants for detailed instructions on how to fill out the budget section.)

Total requested budget: \$457,365

#### Year 1

| Description                                                 | ARC     | Admin Org |
|-------------------------------------------------------------|---------|-----------|
|                                                             | Cash    | Cash      |
| Total                                                       | 153,927 | 131,984   |
| Personnel                                                   | 125,351 | 128,984   |
| Dr Philipp Bayer (Discovery Early Career Researcher Award)  | 104,316 | 39,850    |
| UWA-funded PhD stipend                                      |         | 35,000    |
| Scholarship for International Research Fees for PhD student |         | 54,134    |
| Research Assistant, 0.2 FTE, level 6                        | 21,035  |           |
| Equipment                                                   | 10,371  | 1,500     |
| Bioinformatics work server                                  | 10,371  |           |
| PhD support package - PC                                    |         | 1,500     |
| Maintenance                                                 | 18,205  | 1,500     |
| General lab consumables                                     | 1,500   |           |
| DNA extraction kits                                         | 1,453   |           |
| Oxford Nanopore library preparation                         | 4,834   |           |
| Oxford Nanopore PromethION sequencing (AGRF)                | 5,348   |           |
| Illumina sequencing (Garvan Institute)                      | 5,070   |           |
| PhD support package - consumables                           |         | 1,500     |

#### Year 2

| Description                                                 | ARC     | Admin Org |
|-------------------------------------------------------------|---------|-----------|
|                                                             | Cash    | Cash      |
| Total                                                       | 152,814 | 140,236   |
| Personnel                                                   | 125,749 | 136,886   |
| Dr Philipp Bayer (Discovery Early Career Researcher Award)  | 104,316 | 47,753    |
| UWA-funded PhD stipend                                      |         | 35,000    |
| Scholarship for International Research Fees for PhD student |         | 54,133    |
| Research Assistant, 0.2 FTE, level 6                        | 21,433  |           |
| Travel                                                      | 4,700   | 1,850     |
| PAG conference for CI Bayer                                 | 4,700   |           |
| PhD support package - GRS travel award                      |         | 1,850     |
| Maintenance                                                 | 18,205  | 1,500     |

|                                              |       |       |
|----------------------------------------------|-------|-------|
| General lab consumables                      | 1,500 |       |
| DNA extraction kits                          | 1,453 |       |
| Oxford Nanopore library preparation          | 4,834 |       |
| Oxford Nanopore PromethION sequencing (AGRF) | 5,348 |       |
| Illumina sequencing (Garvan Institute)       | 5,070 |       |
| PhD support package - consumables            |       | 1,500 |
| Other                                        | 4,160 |       |
| Open Access Publication Fees                 | 4,160 |       |

### Year 3

| Description                                                 | ARC     | Admin Org |
|-------------------------------------------------------------|---------|-----------|
|                                                             | Cash    | Cash      |
| Total                                                       | 150,624 | 146,552   |
| Personnel                                                   | 126,147 | 145,052   |
| Dr Philipp Bayer (Discovery Early Career Researcher Award)  | 104,316 | 55,919    |
| UWA-funded PhD stipend                                      |         | 35,000    |
| Scholarship for International Research Fees for PhD student |         | 54,133    |
| Research Assistant, 0.2 FTE, level 6                        | 21,831  |           |
| Travel                                                      | 9,150   |           |
| PAG conference for CI Bayer                                 | 4,700   |           |
| PAG conference for PhD student                              | 4,450   |           |
| Maintenance                                                 | 11,167  | 1,500     |
| General lab consumables                                     | 1,500   |           |
| DNA extraction kits                                         | 1,453   |           |
| Oxford Nanopore library preparation                         | 4,834   |           |
| Illumina sequencing (Garvan Institute)                      | 3,380   |           |
| PhD support package - consumables                           |         | 1,500     |
| Other                                                       | 4,160   |           |
| Open Access Publication Fees                                | 4,160   |           |

## E2. Justification of funding requested from the ARC

*(Fully justify, in terms of need and cost, each budget item requested from the ARC. Use the same headings as in the Description column in the Budget Table of this application. (Upload a PDF of up to four A4 pages and within the required format))*

### Budget Justification

Uploaded PDF file follows on next page.

## E2. Justification of funding requested from the ARC

I request funding for three years. High-performance computing facilities are available via the Pawsey Supercomputing Centre and laboratory and greenhouse facilities are available via the University of Western Australia.

### PERSONNEL

**Dr Philipp Bayer (Discovery Early Career Researcher Award):** I will be leading the project, ensuring that all work is completed is on schedule, that all students involved will receive appropriate supervising, and that all analysis steps are carried out to my satisfaction. The ARC base salary for a DECRA candidate is \$104,316 per year.

**Research Assistant, 0.2 FTE, level 6:** I request funds for a level 6 Research Assistant (RA) to undertake the laboratory work required for this project. This project requires plant growth, DNA extraction and library preparation of Brassicaceae genomes. As I am a purely computational researcher I will require an RA to carry out the laboratory work for the data generation. The RA will need to be experienced in greenhouse work, as well as DNA extraction and library preparation. In particular, the long-read sequencing steps outlined here require significant laboratory experience as the DNA extracted needs to be of high molecular weight. This will require the RA to establish novel DNA extraction methods in-house. The selected RA is based in the Batley research group as Prof. Batley provides a laboratory and access to greenhouses containing the latest equipment necessary for plant growth and library preparation. This would support the ongoing collaboration with the Batley group supporting data generation and wet-lab validation of my computational genomics research.

### TRAVEL

**PAG conference for CI Bayer:** Plant genomics is a rapidly moving field. I request funds to attend the Plant and Animal Genome conference in San Diego every January. This is the primary event for plant genome researchers with more than 3000 attendees annually. I have presented a talk and a poster at almost every PAG since I started my PhD. It is also the conference where international plant genomics consortia hold their yearly general meeting. I participate and present in the Multinational *Brassica* Genome steering committee and the Wheat Information System (WheatIS) meetings of the two bodies coordinating any Brassicaceae sequencing efforts and wheat information exchange and storage systems. These meetings attract the leading researchers in the Brassicaceae and wheat genomics communities, including many comparative genomics researchers. The meetings are a very important platform for me to present my work and to establish and nurture international connections. I plan to travel to PAG in year 2 and 3. In year 2 I will give present the Brassicaceae-wide comparative genomics results. In year 3 I will present the results of the gene loss analysis. Costs are based on a return economy airfare between Perth and San Diego of \$2,500, registration fee of \$700 and subsistence and accommodation for 5 days at \$300 per day. The total for this trip is \$4,700 in the second and the third year of this project.

**PAG conference for PhD student:** The PhD student will join me in presenting at the Plant and Animal Genome Conference where they will present their work. It is important for the student's career to present their results to the international research community to gain experience presenting and defending their research, as well as to make connections with the international research community and potential future employers. The total costs are based on a return economy airfare between Perth and San Diego of \$2500, a student registration fee of \$450 and subsistence and accommodation for 5 days at \$300 per day. The total for this trip will be \$4450 in year 3. Additional funds will be sourced from UWA to support travel for national conferences.

### EQUIPMENT

**Bioinformatics work server:** Genomics and bioinformatics research is extremely computationally intensive. The datasets studied are very large which leads to memory-intensive applications. The main computational hardware required for this project will be supplied by the Pawsey Supercomputing Centre in collaboration with UWA. As the Pawsey Supercomputing Centre employs a queueing system it can take several hours before submitted code starts to run. This can seriously hamper the development of new or experimental



code, as the code may require several submissions, as early iterations may still contain hidden errors that will lead to the code crashing on the supercomputer. Therefore, I request to buy one bioinformatics work server which will enable me and the PhD student to go through fast iterations when developing our code.

I request \$10,371 in year 1 to purchase a Dell Precision R740 server (5 TB HDD, 128 GB RAM) which will be used to develop and optimise code which will be applied for the subsequent analysis of the large genomic datasets on the Pawsey Supercomputing Centre.

## MAINTENANCE

**General lab consumables:** These general lab consumables contain everyday disposables such as gloves, glassware, plasticware, and general laboratory costs, but also to pay for any greenhouse costs as part of the RA's use of the Batley lab's glasshouses. The RA will require these consumables to grow plants, extract DNA and prepare libraries for genome sequencing.

**DNA extraction kits:** I will resequence eight Brassicaceae genomes (*A. alpina*, *A. arabicum*, *C. hirsuta*, *L. meyenii*, *R. raphanistrum*, *R. sativus*, *T. hassleriana*, and *T. arvense*) and use this data to assemble high-quality genome assemblies to improve on the already existing Illumina-only assemblies for these plants. I will also sequence four wild *Brassica* relatives (*B. insularis*, *B. montana*, *B. rupestris*, and *B. villosa*) as these genomes have not been assembled before, yet these wild species represent valuable sources of novel genes, especially disease resistance genes (*R*-genes). The high quality of these genomes to be assembled will ensure that the gene birth and gene death models are as accurate as possible since the probability of misassemblies and missed genes will be as minimal as currently possible.

Seeds of all plants to be sequenced are available from the laboratory of Professor Batley or international collaborators, for example, the large Brassicaceae collections of Prof. Chris Pires at the University of Missouri. The seeds will be grown in collaboration with the Batley lab at UWA. For subsequent library preparation steps, the RA will extract DNA from the greenhouse plants. As I propose to use Oxford Nanopore Technologies® I require high molecular weight DNA, which requires special DNA extraction kits. For this task, I have chosen the Qiagen DNeasy PowerClean Pro Clean Up Kit with two Nanobind Plant Nuclei Big DNA Alpha Kits for a total of \$1,453 per year. This will ensure that all extracted DNA is of sufficient molecular weight for long-read sequencing.

**Oxford Nanopore library preparation:** Due to the complex and repetitive nature of plant genomes third-generation sequencing technology such as PacBio Sequel II or Oxford Nanopore Technologies® PromethION data is needed to accurately assess any plant genome's transposable element and genetic content. In planning this project, I have attempted to reduce any costs associated with this project. A PacBio Sequel II cell costs around \$1,400 while an Oxford Nanopore Technologies® flowcell costs around \$700-900. For price reasons, I have chosen the Oxford Nanopore Technologies® PromethION as the third generation sequencing machine to be used in this project. The RA will be carrying out the library preparation and will, therefore, need library preparation kits. For the PromethION sequencing, we will use four SQK LSK109 nanopore library preparation kits with the NEBNext® Companion Module for Oxford Nanopore Technologies® Ligation Sequencing for \$4,834 per year.

**Oxford Nanopore PromethION sequencing (AGRF):** The genomes will be assembled using third-generation sequencing technology to ensure accurate comparison of genomes of equivalent quality. Oxford Nanopore currently offers the most cost-effective long read data via the PromethION system, however, this will be reviewed when the data is to be generated.

The Australian Genomics Research Facility (AGRF) charges \$1,337 per PromethION flowcell. I will need 5 flowcells in years 1 and 2, and 2 flowcells in year 3 for a total of 12 flowcells, one per species to be sequenced. This will result in around 90 GB of data per species which is more than enough for a high-quality assembly.

**Illumina sequencing (Garvan Institute):** Due to random deletions in Oxford Nanopore Technologies® PromethION reads, the resulting assemblies need to be polished using Illumina reads. For the species where I plan to reassemble genomes, there is already publicly available Illumina data. However, in most cases, this is older data from early iterations of Illumina machines such as the GenomeAnalyzer I. These older machines have higher error rates than modern iterations such as the NovaSeq 6000. I need to ensure that all assembled genomes have an error profile that is as similar as possible to each other in order not to introduce

any accidental bias, such as it would be introduced if different assemblies would have been polished using different generations of Illumina data.

Garvan charges \$1,690 for one Illumina run which results in approximately 100 to 200 GB of data. For polishing of Nanopore data using Illumina data, it is recommended to use 30x to 50x genome coverage. Brassicaceae genomes have a size between 0.6 GB to 1.2 GB. For twelve species to be sequenced this means that the total genome size of all genomes is from 7.2 GB to 14.4 GB. For the required genome coverage of 30x to 50x that means we require between 216 GB to 432 GB (30x) and 360 GB to 720 GB (50x). That is the equivalent of four Illumina runs. I require two Illumina runs in year 1 and 2 respectively (\$5,070) to sequence 50x Illumina data for all twelve individuals.

## **OTHER**

**Open Access Publication Fees:** With the growth of the open-access publication funding model, there has been an increased cost burden associated with such prolific publication. To offset this cost and enable a greater number of our manuscripts to be published under open access arrangements, I request \$4160 in each of the second and the third years towards the cost of open access publication based on the open-access fees of Plant Biotechnology Journal (US\$2,800, AUD\$4,160), a journal I regularly publish in.

### **E3. Details of non-ARC contributions**

*(Provide an explanation of how non-ARC contributions will support the proposed application. Use the same headings as in the Description column in the above Budget Table of this application. (Upload a PDF of up to two A4 pages and within the required format))*

#### **Details of Non-ARC Contributions**

Uploaded PDF file follows on next page.

### E3. Details of non-ARC contributions

#### PERSONNEL

**Dr Philipp Bayer (Discovery Early Career Researcher Award):** ARC will be providing my base salary, but as I am already paid on level B2 UWA has agreed to pay the gap between the ARC base salary and my salary, which will be level B3 at the start of the DECRA in 2021. Together with on-costs this comes to \$39,850 in year 1, \$47,753 in year 2, and \$55,919 in year 3.

**UWA-funded PhD stipend:** This project is an opportunity to train a PhD student in state-of-the-art comparative genomics, pangenomics, and general bioinformatics techniques. Therefore, I propose to train one applied bioinformatics PhD student who will carry out purely computational research. The student trained by this project will acquire skills and expertise which will ensure a flourishing career within research.

The student will work on several aspects of all three aims described in the project description. The student will work on the genome assemblies and gene annotations of all high-quality Brassicaceae assemblies proposed here. This will teach the student how to use high memory high-performance computing, a valuable and highly sought after skill in the life sciences, where high-performance computing has only been adapted over the last ten years. This will also teach the student how to critically assess genome assemblies and sets of predicted genes as misassemblies (misplaced regions in the assembly) and mis-annotations (false positive gene predictions or missed genes) are still common, even in third-generation sequencing projects. Assembling and annotation this many genomes from across the two Brassicaceae lineages will also be an opportunity for the student to learn about the biology and the life-cycle of plant genomes including amphidiploidisation as it occurred in *Brassica napus*, whole-genome duplications or triplications as it occurred in the ancestors of all Brassicaceae, and the many minor ways genomes change over time in response to environmental factors. The genome assemblies in light of their respective evolution will be an opportunity for the student's first publication.

The student will also work on the machine learning part of this project. Machine learning tools have joined the toolkit of industry and across almost all areas of science due to the large amount of novel data being generated due to the ubiquity of computing. This makes skills in critical usage of machine learning highly valuable and therefore desirable for PhD graduates to be competitive in the job market. As part of the second aim, the student will work on the critical evaluation of different machine learning prediction algorithms such as XGBoost, Lasso, Random Forests, and even Artificial Neural Networks to improve the prediction of gene variability. This will require the student to learn how to critically evaluate these models and how to employ different approaches to interpretable machine learning to learn what the models have learned exactly. This critical evaluation will be an opportunity for the student's second publication.

The student will also work on the third aim, the modelling of gene variability in regards to different subclasses of genes. This provides an opportunity for the student to learn how to discover different classes of genetic elements and run large-scale gene class computational discovery pipelines requiring many CPUs over many nodes in a high-performance computing environment. This will further the student's knowledge of the effect of the environment and the organism's history on the size and organisation of its genome. How these different classes grow and shrink in different Brassicaceae lineages, and how this ties in with different environments and genomic events such as whole-genome duplications, provides an opportunity for the student's third publication.

The planned publications described here will be accompanied by the student writing relevant review publications and book chapters for appropriate review journals such as Trends in Genetics or Current Opinion in Plant Biology.

I currently co-supervise 10 PhD students and have co-supervised three students to completion. I am therefore prepared to become the primary supervisor for my first student as part of the DECRA project. In addition to the student agreed to be funded by UWA as part of this DECRA, I expect to recruit an additional 2-4 PhD students to my group as it becomes established. These students will be recruited through competitive scholarships. These students will work on related projects broadening the training outcomes of this funding.

UWA has agreed to fund one student for 3 years for \$30,000 per year, including a \$15,000 6 month extension. In the budget, this extension has been spread out over all three years.

**Scholarship for International Research Fees for PhD student:** So far in my career in Perth I have struggled to attract local students for a PhD position. This is partially due to Perth being one of the most remote large cities on the planet, but also due to there being no local bioinformatics MSc or BSc course, so there is no pipeline from which local students could be recruited. Out of the ten PhD students I am currently supervising seven are international students. It is therefore likely that the PhD student chosen for this project is an international student which will require fees to be paid to UWA. In this case, UWA has agreed to pay for the student's international research fees. This will ensure that I can attract highly talented students from overseas. As with the UWA-funded PhD stipend, there is a 6-month extension included in the budget which has been spread out over all three years. The value of the scholarship is \$54,134 per year.

## TRAVEL

**PhD support package – GRS travel award:** The UWA Graduate Research School provides each student with up to \$1,850 for travel. I aim to send the PhD student to the COMBIO conference in Melbourne. COMBIO is the leading genomics conference in Australia. This is a premium opportunity for the student to present their research to the national genomics community and start to build collaborations. Student registration at the COMBIO conference is \$425. Return flights from Perth to Melbourne are approximately \$500. A 5-night hotel stay in Lorne comes to roughly \$500 for a total conference cost of \$1,425.

## EQUIPMENT

**PhD support package – PC:** The PhD student will be provided with one PC via the UWA Graduate School so that the student can carry out their computational analyses.

## MAINTENANCE

**PhD support package – consumables:** The UWA Graduate Research School provides each student with \$1,500 for consumables per year. These consumables will be used as contribution to the genome sequencing projects by sequencing additional Brassicaceae wild lines as we receive them from international collaborators.

In addition to the direct costs listed in the budget table, and any salary shortfall, The University of Western Australia will also provide the indirect costs associated with the project.

## Part F - Participant Details including ROPE (Dr Philipp Bayer)

### F1. Personal Details

(To update any Personal Details, click on the 'Manage Personal Details' link below. Note this will open a new browser tab. When returning to the form ensure to 'Refresh' the page to capture the changes made to the DECRA candidate's profile.

Note: The Phone Number, Date of Birth, Country of Birth and Indigenous Status parts of the question and corresponding answers will not appear in the PDF version of the form.)

Participation Type

Discovery Early Career Researcher Award

Title

Dr

First Name

Philipp

Family Name

Bayer

Citizenship

Germany

Australian Permanent Resident

Yes

Australian Temporary Resident

N/A

### F2. Current country of residence

(If the DECRA candidate is a Foreign National, they must obtain a legal right to work and reside in Australia.)

Australia

### F5. Qualifications

(To update any qualifications, click on the 'Manage Qualifications' link below. Note this will open a new browser tab. When returning to the form ensure to 'Refresh' the page to capture the changes made to the DECRA candidate's profile.)

| Conferral Date | AQF Level       | Degree/Award Title   | Discipline/Field | Awarding Organisation    | Country of Award |
|----------------|-----------------|----------------------|------------------|--------------------------|------------------|
| 04/05/2016     | Doctoral Degree | Doctor of Philosophy | Bioinformatics   | University of Queensland | Australia        |
| 11/08/2009     | Bachelor Degree | Bachelor of Biology  | Life Sciences    | University of Muenster   | Germany          |

### F6. Research Load (non-ARC Grants and Research)

*(Provide details of research funding from non-ARC sources (in Australia and overseas). For research funding from non-ARC sources, list all projects/applications/awards/fellowships awarded or requests submitted involving that DECRA candidate for funding for the years 2020 to 2024 inclusive.)*

Uploaded PDF file follows on next page.

Funding from non-ARC sources

| <b>Description</b><br>(all named investigators on any proposal or grant/ project/ fellowship in which a Participant is/was involved, project title, source of support, scheme and round) | <b>Same Research Area</b> (Yes/No) | <b>Support Status</b><br>(Requested/Current/Past) | <b>Application/ Project ID</b><br>(for NHMRC applications only) | <b>2020</b><br>\$'000 | <b>2021</b><br>\$'000 | <b>2022</b><br>\$'000 | <b>2023</b><br>\$'000 | <b>2024</b><br>\$'000 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|---------------------------------------------------|-----------------------------------------------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|
| <b>Dr. Philipp Bayer</b> – Forrest Fellowship – Forrest Research Foundation                                                                                                              | Y                                  | C                                                 | n/a                                                             | 18                    | 0                     | 0                     | 0                     | 0                     |



## F7. Currently held ARC Projects

*(This information is auto-populated. If you have any concerns with the information recorded here, please contact your Administering Organisation's Research Office.)*

## F8. What will the DECRA candidate's time commitment (percentage of time) be to research activities related to this project?

*(Enter the percentage of time the DECRA candidate will be committing to the project. The number must be a whole number only without the percentage character.)*

0.9

## F9. Eligibility - Relevant Qualification

*(Please select the qualification which is most relevant to the application)*

| Degree/Award Title   | Awarding Organisation    | Conferral Date |
|----------------------|--------------------------|----------------|
| Doctor of Philosophy | University of Queensland | 04/05/2016     |

## F10. Eligibility - Has the DECRA candidate been granted an extension by the Administering Organisation, to the eligibility period due to a significant career interruption as outlined in the grant guidelines?

*(If the DECRA candidate's qualification relevant to this application (listed in question F9) was awarded prior to 1 March 2015 and they have had a significant career interruption (as listed in the grant guidelines), the DECRA candidate will need to seek an extension to the eligibility period through their Deputy Vice-Chancellor (Research).)*

No

## F11. Eligibility - Select the category of career interruption claimed (more than one may be selected)

*(Choose all types of career interruptions which have been claimed and granted by the DECRA candidate's Deputy Vice-Chancellor (Research).)*

*Select a type of interruption and click 'Add'.)*

## F12. Eligibility - What is the total period of extension that the DECRA candidate has claimed?

*(Select the period of time which most closely equals the total period of extension claimed.)*

## F13. Eligibility - Current Research Fellowship or Award funded by other Australian Government agencies

*(Do not list Fellowships and Awards granted by the ARC. Only list Fellowships and Awards from other agencies.)*

Does the DECRA candidate hold a current Research Fellowship or Award funded by other Australian Government agencies?

No

## F14. Project Relinquishment or Application Withdrawal

*(ARC grant guidelines specify the limits on the number of applications and projects per named participant. Should this DECRA application be successful the DECRA will exceed ARC project limits and must meet the project limits under the grant guidelines before the project can start. Project limits can be met by relinquishing existing active*

project(s), or relinquishing role(s) on existing active projects, or withdrawing application(s) that would exceed the project limits)

|  |
|--|
|  |
|--|

**F15. Research Opportunity and Performance (ROPE) - Current and previous appointment(s) / position(s) - during the past 10 years**

(To update any details in this table, click on the 'Manage Employment Details' link in this question. Note this will open in a new browser tab. 'Refresh' the application page when returning to the form to capture changes made to the DECRA candidate's profile.)

| Description             | Department                    | Contract Type | Employment Type | Start Date | End Date   | Organisation                        |
|-------------------------|-------------------------------|---------------|-----------------|------------|------------|-------------------------------------|
| Forrest Fellow          | School of Biological Sciences | Contract      | Full Time       | 01/01/2018 | 31/01/2021 | The University of Western Australia |
| Postdoctoral researcher | School of Biological Sciences | Contract      | Full Time       | 07/09/2015 | 31/12/2017 | The University of Western Australia |

**F16. Research Opportunity and Performance Evidence (ROPE) - Academic Interruptions**

(You must read the ROPE Statement <http://www.arc.gov.au/arc-research-opportunity-and-performance-evidence-rope-statement> before filling out this section.)

Has the DECRA candidate experienced an interruption that has impacted on their academic record?

|    |
|----|
| No |
|----|

**F17. Research Opportunity and Performance Evidence (ROPE) - Details of the DECRA candidate's academic career and opportunities for research, evidence of research impact and contributions to the field, including those most relevant to this application**

(Provide details of the DECRA candidate's academic career and opportunities. This should not include information presented in the following questions (Upload a PDF of up to five A4 pages))

Uploaded PDF file follows on next page.

## AMOUNT OF TIME AS AN ACTIVE RESEARCHER

I was awarded my PhD four years ago in 2016. I have experienced no academic interruptions and have since been employed at 1.0 FTE.

## RESEARCH OPPORTUNITIES

### 2006-2009 Bachelor of Science, University of Muenster, Germany

In 2009 I was awarded a Bachelor of Science in Life Sciences from the University of Muenster. In my final BSc project in 2009 I worked on differential gene expression in seagrasses under Dr. Jenny Gu and Prof. Erich Bornberg-Bauer, using expressed sequence tags (ESTs) sampled from different depths, later published in *Frontiers in Plant Science* (Dattolo et al., 2013). During this project I learned how to extract biological knowledge from noisy and low-coverage data, skills and knowledge I rely on to this day.

### 2010-2012 Master IT, Bond University, Australia

After my undergraduate degree, I decided that I needed to have a more thorough grounding in programming if I was to progress in applied bioinformatics, so I enrolled for a Master in IT at Bond University. During my final project, I worked on simple artificial neural networks as an alternative approach to genome-wide association studies (GWAS). The project did not significantly increase the accuracy of GWAS, however, this was the very early days of artificial neural networks, and the ANN I built was relatively simple with only three layers. This was my first exposure to data mining and the building of complex machine learning models.

As a side-project during my Master studies Bastian Greshake, I established openSNP. At the time, there was no way for customers of direct-to-consumer genotyping companies such as 23andMe or FamilyTreeDNA to share their data with researchers not affiliated with these companies. A small number of people uploaded their data to their own web-pages or to Github, but they did not annotate the data with phenotypes of interest to researchers. openSNP allows people to share their data under an open license agreement and annotate their data with phenotypes. openSNP is completely open, with the code being hosted on Github under an open license from the start. Working on openSNP taught me many things that have had an impact on my career: how to host large amounts of data in relational databases in an efficient and fast manner, how to design websites, how to maintain computational infrastructure, how to work with and inform stakeholders, and how to attract funding: the site is now funded by user donations via patreon.com.

### 2012-2015 PhD Applied Bioinformatics, Prof. Dave Edwards lab, University of Queensland, Australia

At the end of my Master degree I felt confident enough in my programming skills to join a dedicated bioinformatics research group. I joined the group of Prof. Edwards at the University of Queensland based on his established track record, his access to world-class facilities and the fact that we work well together on a personal level. The Edwards lab is run in close collaboration with the lab of Prof. Jacqueline Batley. Both Profs Batley and Edwards are internationally respected researchers who have been my mentors during my scientific career since, and I have always relied on their advice and support.

In my PhD I primarily worked on developing bioinformatics approaches for high throughput genotyping by skim-based genotyping by sequencing. At the start of my PhD, DNA sequencing was still expensive, so most genotyping applications of were either performed with SNP genotyping arrays or by using reduced representation sequencing such as restriction-site associated DNA marker sequencing (RADseq). In my PhD I developed skim-based genotyping by sequencing, which uses low coverage DNA sequencing of the entire genome to greatly increase the coverage and the number of genetic variants called, though with imputation required to define haplotypes (Bayer et al., 2015). This approach has been adopted world-wide by researchers and breeding companies, including Bayer CropScience, now BASF. I applied skimGBS to identify genetic variants associated with agronomic traits, with one such study leading to the breeding of drought tolerant chickpea varieties currently being grown in Australia and India. In addition, I applied it to map gene conversions and genetic crossovers, and to assess the quality of genome assemblies, leading to major publications in collaboration with international consortia. For example, I used skimGBS to identify errors in the canola genome assembly which was published in *Science* (Chalhoub et al., 2014) and *Nature Genetics* (Kreplak et al., 2019), and similar work has been undertaken in chickpea, wheat, pea and lentil.

While I was working on skimGBS I also contributed to numerous other projects. Together with Paul Visendi, I assisted in the assembly of the wheat chromosome arm 7DS, which recently published as part of the complete wheat genome assembly in *Science* (Appels et al., 2018). Other projects included the analysis of

an interspecific crossing dataset to map meiotic between *Brassica* chromosomes (Mason et al., 2014) and to map centromere locations (Mason et al., 2016), as well as large-scale identification of genomic variants on the group 7 chromosomes of wheat (Lai et al., 2015). I published two book-chapters around skimGBS (Bayer et al., 2106b) and salinity (Bayer et al., 2016a). Together with Dr. Kenneth Chan I was the system administrator for the three work-servers and several web-servers hosted by the Edwards group. This work taught me a huge amount about the structure of bioinformatics work environments, which now allows me to accurately evaluate what my work requires in terms of computational infrastructure.

The research during my PhD was undertaken in close collaboration with Bayer CropScience (now BASF), and during my PhD, I had the opportunity to spend time with Bayer researchers in Ghent, Belgium. I initially stayed four weeks to work on joint projects with Bayer researchers and to exchange data and results from my skim-based genotyping by sequencing approach. I have since returned to Bayer CropScience two more times to work on different projects, and I meet with BASF collaborators once a year at the Plant and Animal Genome conference in San Diego. The projects I carried out with Bayer CropScience are confidential and commercially sensitive so cannot be published.

#### 2015-2017 Postdoctoral researcher, Prof. Dave Edwards lab, University of Western Australia, Australia

At the end of my PhD, the Edwards group moved to the University of Western Australia. Due to my experience I was invited to help set up the new location as a full-time postdoctoral researcher. At this point I also decided to extend my research to the field of pangenomics, approaches which characterise the full extent of genetic variation of a species. We were the first researchers to use this approach in plants, and remain international leaders in the field which is now being applied to many crop and wild plant species.

The first publication resulting from this work was the first pangenome for a plant species, led by my PhD student Agnieszka Golicz, and in collaboration with the international *Brassica* genome community (Golicz et al. 2016). I extended my pangenome research to study agronomically important disease resistance genes, which led to the publication of an atlas of disease resistance genes in *B. oleracea* (Bayer et al., 2018). In characterising the diversity of disease resistance genes, irregularities in gene numbers in some *Brassica* assemblies led me to discover that some computational gene prediction pipelines accidentally remove resistance genes from their results. This finding was recently published in *Nature Plants* (Bayer et al., 2018).

Working with diverse international teams and consortia is an important aspect of applied bioinformatics research and I continue to work on establishing and maintaining international and national collaborations. In 2016 I visited Rothamsted Research to work on QTLNetMiner (now KNetminer), a knowledge graph-based web application which is used to visualise and explore links between genes and the published literature. In my one-week visit to learn about how the system works, I established an instance hosted at UWA and wrote the entire setup documentation which is still in use, almost completely unchanged.

Between 2016 and 2017 I accepted the position of the Western Australia representative for COMBINE, the student arm of the Australian Bioinformatics and Computational Biology Society (ABACBS). In that role, I helped organise 3 workshops and speaking events for early career and PhD bioinformatics researchers around Perth. This experience has helped me establish connections with life science computational researchers in Western Australia.

Since 2016, I joined EMBL-ABR (European Molecular Biology Laboratory – Australian Bioinformatics Resource) as joint head of the UWA node with Prof. Dave Edwards. EMBL-ABR is a national initiative bringing bioinformatics researchers from 13 universities and research institutes together to coordinate and streamline the development of bioinformatics data, tools, standards, platforms, training, and compute infrastructure. Activities are coordinated through an annual all-hands meeting and monthly online calls. Being a head of the UWA EMBL-ABR node has given me insight into how large-scale science is coordinated and carried out on a national level.

#### 2018-current Forrest Fellowship, Prof. Dave Edwards lab, University of Western Australia, Australia

In the second year of my first postdoctoral appointment, I applied for a Forrest Fellowship with the Forrest Research Foundation, which was awarded and started in 2018. I was the third-highest rated candidate, which means that I could only be awarded a non-stipendiary fellowship which pays for living costs in Forrest Hall, newly built living accommodation next to UWA. Due to my outreach work, the foundation asked whether I want to become a 'senior fellow', which means that I spend time online and in the afternoons building a community involving all Forrest Foundation postdoctoral researchers and PhD students. This position has enabled me to get in contact with billionaire philanthropists, which has taught me what kind of projects

philanthropic foundations are looking for when allocating funds, as well as how to make my voice heard when it comes to steering the direction of philanthropic efforts.

During this time I worked on several comparative genomics publications. I helped assemble hundreds of barley exomes and searched them for *Restorer-of-fertility*-like loci (Melonek et al., 2019). I analysed and compared the genetic content of three newly published soybean genomes (Valliyodan et al., 2019). I contributed my expertise in comparative genomics to the mapping of genetic variability and associated one indel with flowering time in lupins (Taylor et al., 2019a, Taylor et al., 2019b). I wrote a paper searching for variance in disease resistance genes in the previously published *B. oleracea* pangenome and showed that variability in these genes is driven by proximity to transposable elements (Bayer et al., 2019). I also published my first publication as the corresponding author: an introductory review on genome-wide association studies in plants (Anderson et al., 2019). These publications are signifying my change to an independent bioinformatics researcher outside the Edwards lab. During this time I also became the father of two sons now aged 1 and 3 which requires occasional restrictions on attending international meetings or travel.

As a Forrest Fellow, I continued to work in the Edwards lab in a more independent role. I continued working on pangenomes of crops, especially Brassicaceae. I assembled three pangenomes for *Brassica oleracea* and *B. rapa*, and their amphidiploid progenitor *B. napus* (canola). This is the largest pangenome ever produced which gives us unprecedented insight into the history of canola. In this work, I show how genetic variety was lost or was never introduced during the breeding process in canola and I used modelling and machine learning to show which genomic elements are linked with variable genes.

I also continued my work on mining genome assemblies for novel disease resistance genes. I worked with the PhD student Soodeh Tirnaz on mining all published Brassicaceae genomes for novel disease resistance genes, discovering hundreds of previously unknown resistance genes. This work has undergone peer review and we just resubmitted the revised manuscript.

In 2019 I was nominated for the Woodside Early Career Scientist of the Year Prize (\$10,000) as part of the WA Premier's Science Awards. I was one of the four finalists from all across Western Australia.

In 2018 my first co-supervised PhD Mahsa Mousaviderazmahalleh student successfully completed her studies, and two MSc students I co-supervised graduated. In 2019 two of my co-supervised PhD students (Armin Scheben and Yuxuan Yuan) graduated and are now working as postdocs at Cold Spring Harbor Laboratory and at the Chinese University of Hong Kong. I am currently a 20% and coordinating supervisor for 7 PhD students and 3 MSc students with 2 PhD students and all 3 MSc students expected to submit their theses in 2020. I share my office space with all of my students which means that I am usually the first point of contact for all questions and problems, a task which I greatly enjoy.

## Teaching

Since 2015 I have contributed to teaching SCIE4002, an introduction to bioinformatics for Biology MSc students. I teach one-third of the course, designing it to make it as interactive as possible, providing valuable practical skills. The course is popular with students, with increasing numbers each year reflecting the growing importance of these skills to biologists, and in 2018 the course won an award for having received high student evaluation marks for three years in a row. In 2018 I won the UWA Student Guild's Students' Choice award for my teaching of SCIE4002.

I am keen to share my expertise and train others, this also helps me build and extend my own skillset, so during my PhD, I trained to become a certified Software Carpentry and later, Data Carpentry instructor. The certification course includes a four-week online workshop on the psychology and effectiveness of teaching programming. The certificate allows me to teach programming workshops where I teach software and data skills to scientists from diverse backgrounds. Since this training in 2014, I have taught seven Software Carpentry and two Data Carpentry workshops, and have organised the first-ever Data Carpentry workshop at UWA. Becoming a Carpentries instructor has greatly improved my tertiary education teaching skills, and has provided valuable insights into the psychology of effective teaching and learning.

During 2017 I was a mentor for Mozilla Open Science, which runs short yearly courses where attendants where mentors who are running successful open source or open science projects share their experience with others who are just starting their open source project. The focus of the program is to help grow communities around open source projects. In that capacity, I mentored two projects – one on creating simple programming instructions for scientists, and one on creating materials for research and writing sprints held

at universities. Working with Mozilla Open Science has taught me how to effectively and empathetically mentor students and researchers.

In 2017 I founded the UWA Hacky Hour. Hacky Hours are regular meetings around Australian research universities which are intended to help researchers having problems with code or data, or just want to chat possible approaches for solutions to their computational problems. Before Hacky Hour, there was no similar institution lending computational help to researchers struggling with their analyses. The Hacky Hour at UWA has grown into a small regular community with 4 to 7 people attending every week and roughly 3 problems solved per week, for a total of roughly 200 problems solved so far for the UWA academic community.

### **Outreach**

In 2017 I joined the School of Biological Science's Community and Outreach committee. In that position, I had a hand in creating the school's social media strategy. I started the school's first Twitter account and took over running the school's Facebook account. Using these channels I regularly disseminate new press releases and publications and advertise for events, such as public presentations, by the school's researchers or by affiliated schools. This work has taught me how to present my research so that the interested public can understand it better.

In this time I became a regular attendant of the weekly Perth Machine Learning Group. This is a meetup hosted and maintained by Perth industry professionals who use machine learning in their work. There are no regular attendants from academia except for me, instead, the majority of attendants work in the mining, energy, and finance industry. Regular attendance in this group has taught me everything I know about modern machine learning and deep learning approaches and best practices followed in the industry. The meetup group has also put me in contact with countless industry professionals, mostly from Western Power and Perth-based mining start-ups. I have presented my gene space deletion modelling work to this community in order to receive technical feedback on my approaches which allowed me to access industry state-of-the-art approaches for my work.

### **RESEARCH ACHIEVEMENTS AND CONTRIBUTIONS**

All awards are sorted chronologically, with the newest first.

I regularly review manuscripts for scientific journals. I have reviewed manuscripts for *Plant Biotechnology Journal* (3 manuscripts), *Scientific Reports* (1), *Nature Plants* (1), *Molecular Biology and Evolution* (1), *Current Opinion in Plant Biology* (1), *BMC Genomics* (1), *Agronomy* (1), *G3: Genes|Genomes|Genetics* (1), *The Plant Genome* (1), *International Journal of Molecular Sciences* (1), *Plant and Soil* (1), *Scientific Data* (1), and *Bioinformatics* (1). I am an editor for the journal *Agronomy*.

### **Academic output**

I have always strived to be at the forefront of bioinformatics research. I have been involved in pioneering research in skim-based genotyping by sequencing, and I was involved in the very first plant pangenome paper and the biggest and most complex plant genome assemblies in recent years, canola (Chalhoub et al., 2014), wheat (Appels et al., 2019), and pea (Kreplak et al., 2019). I have tried not to focus myself too much on one research niche, as bioinformatics is an extremely fast-moving field. Therefore I have tried to become involved in different fields and study organisms where I could apply my data analysis and logic skills. By working in diverse crops such as pea, wheat, canola, lupin, clover, and seagrasses, and by working on diverse fields such as genome assembly, population genetics, machine learning, database design, QTL-mapping, genome-wide association studies, and more.

At the same time, I believe that knowledge that is not shared is knowledge that is essentially lost. Therefore I have always strived to improve my teaching and started new initiatives to disseminate knowledge and best computational approaches around academia and the public, including the UWA Hacky Hour, Software and Data Carpentries workshops, as well as my teaching of SCIE4002.

### **Delivery to industry**

Under Prof. Edwards, I have always worked closely with Bayer CropScience's canola research department, which was sold to BASF in 2017. Parts of my research into applying skim-based genotyping by sequencing formed the bases for an ARC Linkage Grant with Bayer CropScience (LP140100537) worth \$630,000. During and after my PhD I have visited Bayer CropScience several times to complete industry-relevant tasks. Among the 4 projects I worked on with Bayer CropScience, I am most proud of the identification of the size

of an alien introgression in a plant breeding line. On Monday I received the task and on Friday I submitted the finished report with the size and location of the Introgression. Later work by others with better technology confirmed my findings. In other projects with Bayer CropScience, I worked on assembling genome assemblies of new Bayer CropScience cultivars in canola and wheat, which is still a large-scale effort due to the large resources required by the massive wheat genome. In this work, Yuxuan Yuan, HueyTyng Lee, Robyn Anderson, Dave Edwards and I developed a novel approach to genome assembly which reduces the need for memory by an order of magnitude. This work has not been published as we are evaluating whether we sell the algorithm to BASF.

Among with Prof. Edwards and Prof. Bennamoun from the UWA School of Computer Science we have just successfully submitted tenders for two projects to the GRDC. I am co-PI on both applications. Both projects have a strong industry and agriculture focus. One project proposes to use machine learning to identify plant stresses from satellite and drone images, the other project focuses on the prediction of traits of agronomic interest, such as yield, using genomics data and pangenomes. Both projects have been signed off and have started at the beginning of 2020.

### **Prizes, honours, awards**

- I was one of the four finalists for the Woodside Early Career Scientist of the Year Prize (\$10,000) among the 2019 WA Premier's Science Awards.
- For my teaching of the bioinformatics course SCIE4002 I have received the UWA Student Guild's Students' Choice Awards for Semester 2, 2018, for the quoted reason: "Philipp's motivation has turned Bioinformatics into an exciting and interesting subject."
- I was nominated for the Rising Stars award 2018 along with 12 other early career researchers at UWA.
- With my side-project openSNP, we won the PLOS/Mendeley "Binary Battle" for the software or website that uses PLOS and Mendeley resources in the most innovative way. The prize came with a cash award of USD 10,001 and a USD 1,000 Amazon AWS voucher. The jury included Tim O'Reilly, founder of O'Reilly media, Juan Enriquez, Managing Director of Excel Venture Management, John Wilbanks, chief commons officer at Sage Bionetworks, James Powell, CTO of Thompson Reuters, and Werner Vogels, CTO of Amazon.com.
- At the end of my Master in IT, I was awarded the John Oglethorpe medal for the highest GPA of a graduating student. During my Master I was awarded Top of Class 5 times, Vice-Chancellor's List for Academic Excellence three times, and once the IT Award Academic Excellence.

### **Funding secured**

- In the middle of my first post-doctoral position in 2017, I was awarded an inaugural Forrest Research Foundation non-stipendiary fellowship. This fellowship includes \$20,697 per year to cover rental costs in Forrest Hall, accommodation opened in March 2018 specifically for Forrest scholars and fellows. It includes an additional \$12,000 per year for an appointment to 'senior' fellow whose goal it is to foster a community among inhabitants of Forrest Hall.
- I was awarded a UWA Research Collaboration Award 2017 to spend in 2018. This award entails \$28,100 to pay for sequencing of Western Australian seagrass microbiomes to learn more about nitrogen uptake in these plants.
- During my PhD at UQ in 2014, I was awarded a School of Agriculture and Food Sciences Travel award (AUSD 2000), GRDC travel award (AUSD 1800), which I used to attend the Plant and Animal Genome Conference, the world's largest genomics conference, in San Diego in 2014.
- For my PhD I was awarded a UQ tuition fee and a living costs scholarship for the duration of my PhD.
- For my side-project openSNP I was awarded a Bayer Grants4Apps grant including EUR 5,000, which was used to pay for ongoing server costs.

## F18. Research Opportunity and Performance Evidence (ROPE) - Research Output Context

*(Research context: Provide clear information that explains the relative importance of different research outputs and expectations in the DECRA candidate's discipline/s. The information should help assessors understand the context of the DECRA candidate's academic research achievements but not repeat information already provided in this application. It is helpful to include the importance/esteem of specific journals in their field; specific indicators of recognition within their field such as first authorship/citations, or significance of non-traditional research outputs. (Up to 3750 characters, approximately 500 words).)*

In contrast to many bioinformaticians who provide data analysis support for other researchers resulting in mid-author publications, I strive to lead and publish my research as senior author. I have published 48 peer-reviewed publications and 4 book chapters, I have 2,321 citations, an h-index of 19, and an i10-index of 28 (source: Google Scholar, March 2020). I have 4 first-author publications and 2 first-author book chapters.

I am an author on three publications in Science, the second-highest multidisciplinary science journal (JIF: 41.058). – the first complete wheat genome assembly, where I assisted in the assembly of the 7DS chromosome (IWGSC 2018, 358 citations), the wheat Transcriptome atlas based on the same genome assembly (Ramírez-González et al., 2018, 88 citations) and the first complete canola genome assembly, where I used my skimGBS methodology (Bayer et al., 2016) to find and correct erroneously assembled regions (Chalhoub et al., 2016, 961 citations). I am also an author on one publication in Nature Genetics, the second-highest 'Genetics & Heredity' (JCR® Category) journal with an impact factor of 25.5. This publication introduced the first reference genome of pea, where I developed a novel method based on flow-cytometry sorted reads to assess the genome assembly and fix many errors (Kreplak et al., 2019, 9 citations).

I am the second author on the first pangenome publication in the fifth-highest ranked 'multidisciplinary sciences' (JCR® Category), Nature Communications (JIF: 11.8, Golicz et al., 2016, 99 citations), where I compared and evaluated different assembly approaches. I am the first author on a publication in the third-highest ranking 'plant science' journal, Nature Plants (JIF: 13.3, Bayer et al., 2018, 9 citations), where I showed that some approaches to gene annotation can lead to accidental removal of disease resistance gene candidates.

I am co-author in New Phytologist (8th highest ranked in 'Plant Sciences', IF: 7.4, Mason et al., 2014), and in eight publications (two as first author) in the Plant Biotechnology Journal (9th highest ranked in 'Plant Sciences', IF: 6.8, Lai et al., 2015, Hane et al., 2017, Kaur et al., 2017, Bayer et al., 2017, Hurgobin et al., 2018, Bayer et al., 2019, Dolatabadian et al., 2019, Zhao et al., 2020). I am co-author on four publications in Plant Journal (11th highest ranked, IF: 5.7, Montenegro et al., 2017, Scheben et al., 2018, Melonek et al., 2018, Valliyodan et al., 2019).

In the field of bioinformatics I am a co-author in Briefings in Bioinformatics (1st highest ranked in 'Mathematical & Computational Biology', IF: 9.1, Schneider et al., 2017), Bioinformatics (4th highest ranked 'Mathematical & Computational Biology', IF: 4.5, Yuan et al., 2017a), and BMC Bioinformatics (9th highest ranked, IF: 2.5, Yuan et al., 2017b).

As for non-traditional research output, I extensively use social media to promote my and our group's research, my main account @PhilippBayer now has 4,340 followers, but I also started and run the Twitter account and the Facebook account of our school (@BioSci\_UWA, 728 followers, facebook.com/BioSciUWA, 1,092 followers/1,034 likes). All of my code is openly available in my GitHub account (github.com/philippbayer). In August 2018 I wrote a text called 'Things a scientist is supposed to know' on GitHub under the open CC-BY-SA license. The tweet announcing the text received 115,383 impressions with 5,462 clicks, 320 retweets, and 705 likes. In 2011 I co-founded openSNP, a web-based repository where customers of direct-to-consumer genotyping companies such as 23andMe openly share and annotate their data. This project now hosts data for more than 6,000 individuals.

## F19. Research Opportunity and Performance Evidence (ROPE) – Research Outputs Listing including Ten Career-Best Research Outputs

*(Provide a list of research outputs relevant to this application categorised under the following headings: Ten career-best research outputs; Authored books; Edited books; Book chapters; Refereed Journal articles; Fully refereed conference proceedings; Additional research outputs (including non-traditional research outputs). CVs and theses should not be included in this list. The DECRA candidate's ten career-best research outputs should not be repeated under subsequent headings. (Up to 100 research outputs).)*

### Research Outputs Listing

Generated research output document follows on the next page



## Ten Career-Best Research Outputs

- [1] \* Bayer, Philipp E, Bhavna Hurgobin, Golicz, Agnieszka A., Chan, Chon-Kit Kenneth & Yuxuan Yuan et al. 2017, 'Assembly and comparison of two closely related *Brassica napus* genomes', *Plant Biotechnology Journal*, vol. 15, no. 12, pp. 1–9, doi:10.1111/pbi.12742 (Refereed Journal Article)
- [2] \* International Wheat Genome Sequencing Consortium (IWGSC), Rudi Appels, Kellye Eversole, Catherine Feuillet & Beat Keller et al. 2018, 'Shifting the limits in wheat research and breeding using a fully annotated reference genome', *Science*, vol. 361, no. 6403, doi:10.1126/science.aar7191 (Refereed Journal Article)
- [3] \* Ramírez-González RH, Borrill P, Lang D, Harrington SA & Brinton J et al. 2018, 'The transcriptional landscape of polyploid wheat', *Science (New York, N.Y.)*, doi:10.1126/science.aar6089 (Refereed Journal Article)
- [4] \* Bayer, Philipp E., David Edwards & Jacqueline Batley 2018, 'Bias in resistance gene prediction due to repeat masking', *Nature Plants*, vol. 4, no. 10, pp. 762–765, doi:10.1038/s41477-018-0264-0 (Refereed Journal Article)
- [5] \* B.A. Chalhouh, F. Denceud, S. Liu, I.A.P. Parkin & H. Tang et al. 2014, 'Early allopolyploid evolution in the post-neolithic *Brassica napus* oilseed genome', *Science*, vol. 345, no. 6199, pp. 950–953, doi:10.1126/science.1253435 (Refereed Journal Article)
- [6] \* Golicz, Agnieszka A., Bayer, Philipp E., Barker, Guy C., Edger, Patrick P. & Kim, Hye Ran et al. 2016, 'The pangenome of an agronomically important crop plant *Brassica oleracea*', *Nature Communications*, vol. 7, doi:10.1038/ncomms13390 (Refereed Journal Article)
- [7] Philipp Bayer, P. Ruperao, A.S. Mason, J. Stiller & Kenneth Chan et al. 2015, 'High-resolution skim genotyping by sequencing reveals the distribution of crossovers and gene conversions in *Cicer arietinum* and *Brassica napus*', *TAG. Theoretical and applied genetics. Theoretische und angewandte Genetik*, vol. 128, no. 6, pp. 1039–1047, doi:10.1007/s00122-015-2488-y (Refereed Journal Article)
- [8] \* Jonathan Kreplak, Mohammed-Amin Madoui, Petr Cápál, Petr Novák & Karine Labadie et al. 2019, 'A reference genome for pea provides insight into legume genome evolution', *Nature Genetics*, vol. 51, no. 9, pp. 1411–1422, doi:10.1038/s41588-019-0480-1 (Refereed Journal Article)
- [9] \* Bhavna Hurgobin, Golicz, Agnieszka A, Bayer, Philipp E, Chan, Chon-Kit Kenneth & Soodeh Tirnaz et al. 2018, 'Homoeologous exchange is a major cause of gene presence/absence variation in the amphidiploid *Brassica napus*', *Plant Biotechnology Journal*, pp. 1–10, doi:10.1111/pbi.12867 (Refereed Journal Article)
- [10] \* Babu Valliyodan, Cannon, Steven B, Bayer, Philipp E, Shengqiang Shu & Brown, Anne V et al. 2019, 'Construction and comparison of three reference-quality genome assemblies for soybean', *The Plant Journal*, doi:10.1111/tpj.14500 (Refereed Journal Article)

## Book Chapters

- [1] Robyn Anderson, Cassandra Tay Fernandez, Yuxuan Yuan, Agnieszka A. Golicz & David Edwards et al. 2020, 'Method for Genome-Wide Association Study: A Soybean Example', *Legume Genomics*, Springer US, pp. 147–158, doi:10.1007/978-1-0716-0235-5\_7
- [2] Haifei Hu, Yuxuan Yuan, Philipp E. Bayer, Cassandra T. Fernandez & Armin Scheben et al. 2020, 'Legume Pangenome Construction Using an Iterative Mapping and Assembly Approach', *Legume Genomics*, Springer US, pp. 35–47, doi:10.1007/978-1-0716-0235-5\_3
- [3] Philipp Emanuel Bayer 2016, 'Skim-Based Genotyping by Sequencing Using a Double Haploid Population to Call SNPs, Infer Gene Conversions, and Improve Genome Assemblies', *Plant Bioinformatics*, Springer New York, pp. 285–292, doi:10.1007/978-1-4939-3167-5\_16
- [4] Philipp Emanuel Bayer 2016, 'Genomics of Salinity', *Plant Genomics and Climate Change*, Springer New York, pp. 179–194, doi:10.1007/978-1-4939-3536-9\_9
- [5] Agnieszka A. Golicz, Philipp E. Bayer & David Edwards 2014, 'Skim-Based Genotyping by Sequencing', *Methods in Molecular Biology*, Springer New York, pp. 257–270, doi:10.1007/978-1-4939-1966-6\_19

## Refereed Journal Articles

- [1] Monica Furaste Danilevicz, Cassandra Geraldine Tay Fernandez, Jacob Ian Marsh, Philipp Emanuel Bayer & David Edwards 2020, 'Plant pangenomics: approaches, applications and advancements', *Current Opinion in Plant Biology*, vol. 54, pp. 18–25, doi:10.1016/j.pbi.2019.12.005
- [2] Agnieszka A. Golicz, Philipp E. Bayer, Prem L. Bhalla, Jacqueline Batley & David Edwards 2020, 'Pangenomics Comes of Age: From Bacteria to Plant and Animal Applications', *Trends in Genetics*, vol. 36, no. 2, pp. 132–145, doi:10.1016/j.tig.2019.11.006

- [3] Junliang Zhao, Philipp E. Bayer, Pradeep Ruperao, Rachit K Saxena & Aamir W Khan et al. 2020, 'Trait associations in the pangenome of pigeon pea ( *Cajanus cajan* )', *Plant Biotechnology Journal*, doi:10.1111/pbi.13354
- [4] Robyn Anderson, Philipp E Bayer & David Edwards 2020, 'Climate change and the need for agricultural adaptation', *Current Opinion in Plant Biology*, doi:10.1016/j.pbi.2019.12.006
- [5] Aria Dolatabadian, Philipp E. Bayer, Soodeh Tirnaz, Bhavna Hurgobin & David Edwards et al. 2019, 'Characterization of disease resistance genes in the Brassica napus pangenome reveals significant structural variation', *Plant Biotechnology Journal*, doi:10.1111/pbi.13262
- [6] Mahsa Mousavi-Derazmahalleh, Steven Chang, Geoff Thomas, Mark Derbyshire & Phillip E. Bayer et al. 2019, 'Prediction of pathogenicity genes involved in adaptation to a lupin host in the fungal pathogens Botrytis cinerea and Sclerotinia sclerotiorum via comparative genomics', *BMC Genomics*, vol. 20, no. 1, doi:10.1186/s12864-019-5774-2
- [7] Armin Scheben, Brent Verpaalen, Cynthia T. Lawley, Chon-Kit K. Chan & Philipp E. Bayer et al. 2019, 'CropSNPdb: a database of SNP array data for Brassica crops and hexaploid bread wheat', *The Plant Journal*, vol. 98, no. 1, pp. 142–152, doi:10.1111/tpj.14194
- [8] Hediye Tahghighi, William Erskine, Richard G. Bennett, Philipp E. Bayer & Maria Pazos-Navarro et al. 2019, 'Genetic Diversity Linked to Haplotype Variation in the World Core Collection of Trifolium subterraneum for Boron Toxicity Tolerance Provides Valuable Markers for Pasture Breeding', *Frontiers in Plant Science*, vol. 10, doi:10.3389/fpls.2019.01043
- [9] Salman Alamery, Soodeh Tirnaz, Philipp Bayer, Reece Tollenaere & Boulos Chaloub et al. 2018, 'Genome-wide identification and comparative analysis of NBS-LRR resistance genes in Brassica napus', *Crop and Pasture Science*, vol. 69, no. 1, pp. 72, doi:10.1071/cp17214
- [10] Philipp E. Bayer, Agnieszka A. Golicz, Soodeh Tirnaz, Chon-Kit Kenneth Chan & David Edwards et al. 2018, 'Variation in abundance of predicted resistance genes in the Brassica oleracea pangenome', *Plant Biotechnology Journal*, vol. 17, no. 4, pp. 789–800, doi:10.1111/pbi.13015
- [11] HueyTyng Lee, Agnieszka A Golicz, Philipp E Bayer, Anita A Severn-Ellis & Chon-Kit Kenneth Chan et al. 2018, 'Genomic comparison of two independent seagrass lineages reveals habitat-driven convergent evolution', *Journal of Experimental Botany*, vol. 69, no. 15, pp. 3689–3702, doi:10.1093/jxb/ery147
- [12] Joanna Melonek, Ruonan Zhou, Philipp E. Bayer, David Edwards & Nils Stein et al. 2018, 'High intraspecific diversity of Restorer-of-fertility-like genes in barley', *The Plant Journal*, vol. 97, no. 2, pp. 281–295, doi:10.1111/tpj.14115
- [13] Mahsa Mousavi-Derazmahalleh, Philipp E. Bayer, Bruno Nevado, Bhavna Hurgobin & Dmitry Filatov et al. 2018, 'Exploring the genetic and adaptive diversity of a pan-Mediterranean crop wild relative: narrow-leaved lupin', *Theoretical and Applied Genetics*, vol. 131, no. 4, pp. 887–901, doi:10.1007/s00122-017-3045-7
- [14] Mahsa Mousavi-Derazmahalleh, Bruno Nevado, Philipp E. Bayer, Dmitry A. Filatov & James K. Hane et al. 2018, 'The western Mediterranean region provided the founder population of domesticated narrow-leaved lupin', *Theoretical and Applied Genetics*, vol. 131, no. 12, pp. 2543–2554, doi:10.1007/s00122-018-3171-x
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- [16] Candy M. Taylor, Lars G. Kamphuis, Weilu Zhang, Gagan Garg & Jens D. Berger et al. 2018, 'INDEL variation in the regulatory region of the major flowering time gene LanFTc1 is associated with vernalization response and flowering time in narrow-leaved lupin (Lupinus angustifolius L.)', *Plant, Cell & Environment*, vol. 42, no. 1, pp. 174–187, doi:10.1111/pce.13320
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## Certification

### Certification by the Deputy/Pro Vice-Chancellor (Research) or their delegate or equivalent in the Administering Organisation

I certify that—

- I have read, understood and complied with the *Grant Guidelines for the Discovery Program (2019 edition)*, (grant guidelines) and, to the best of my knowledge all details provided in this application form and in any supporting documentation are true and complete in accordance with the grant guidelines.
- Proper enquiries have been made and I am satisfied that the Discovery Early Career Researcher Award (DECRA) candidate listed in this application meets the requirements specified in the grant guidelines, including having been awarded a PhD on or after 1 March 2015. Where the DECRA candidate has allowable career interruptions, sufficient evidence has been provided to the Administering Organisation and based on this evidence, I certify that the candidate has an award of PhD date together with an allowable period of career interruption (as listed in the grant guidelines) that would be commensurate with an award of PhD date on or after 1 March 2015.
- Where the DECRA candidate holds a research higher degree, that is not a PhD, sufficient evidence has been provided to the Administering Organisation and based on this evidence, I certify that the candidate's qualification meets the level 10 criteria of the *Australian Qualifications Framework Second Edition*.
- I certify that where a DECRA candidate has more than one PhD, the earliest PhD has been selected and meets the eligibility requirements, including having been awarded a PhD on or after 1 March 2015.
- Upon request from the ARC, this organisation will provide evidence to support a career interruption justification in relation to the PhD award date.
- The ARC reserves the right to audit any evidence on which an application is based.
- I will notify the ARC if there are changes to the DECRA candidate after the submission of this application.
- The listed participants are responsible for the authorship and intellectual content of this application, and has appropriately cited sources and acknowledged significant contributions to this application.
- To the best of my knowledge, all Conflicts of Interest relating to parties involved in or associated with this application have been disclosed to the Administering Organisation, and, if the application is successful, I agree to manage all Conflicts of Interest relating to this application in accordance with the *Australian Code for the Responsible Conduct of Research (2018)*, the *ARC Conflict of Interest and Confidentiality Policy* located on the ARC website and any relevant successor documents.
- I have obtained the agreement, attested to by written evidence, of all the relevant persons and organisations necessary to allow the project to proceed. This written evidence has been retained and will be provided to the ARC if requested.
- This application complies with the eligible research requirements set out in the *ARC Medical Research Policy*, located on the ARC website.
- This application does not request funding for the same research activities, infrastructure or project previously funded or currently being funded through any other Commonwealth funding.
- If this application is successful, I am prepared to have the project carried out as set out in this application and agree to abide by the terms and conditions of the grant guidelines and the relevant Commonwealth grant agreement.
- The project can be accommodated within the general facilities of this organisation and if applicable, within the facilities of other relevant organisations specified in this application and sufficient working and office space is available for any proposed additional staff.
- All funds for this project will only be spent for the purpose for which they are provided.
- The project will not be permitted to commence until there is an ethics plan in place to ensure that the appropriate clearances or other statutory requirements will be met before the part/s of the project that require those clearances commence.
- I consent, on behalf of all the parties, to this application being referred to third parties, including to overseas parties, who will remain anonymous, for assessment purposes.

- I consent, on behalf of all the parties, to this application being provided to third parties for the purposes of assessment for potential other funding opportunities.
- I consent, on behalf of all the parties, to the ARC copying, modifying and otherwise dealing with information contained in this application.
- To the best of my knowledge, the Privacy Notice appearing at the top of this form has been drawn to the attention of the DECRA candidate whose personal details have been provided in the Participant section of the application.